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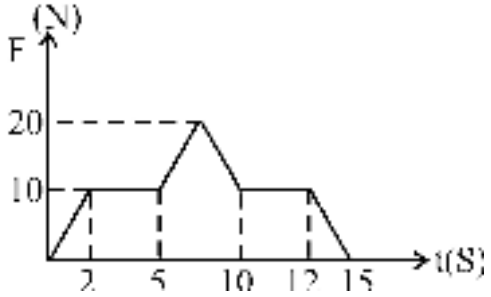
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PHYSICS

1) For the given F-t graph, if force starts acting on a body of mass 2 kg, which is initially at rest, then find its velocity after 15 sec :-



(1) 75 m/s

(2) 150 m/s

(3) 125 m/s

(4) None of these

2) A force $\vec{F} = 3\hat{i} + 4\hat{j} + 12\hat{k}$ Newton produces acceleration of 1m/sec^2 in a body. The mass of the body is (in kg) :-

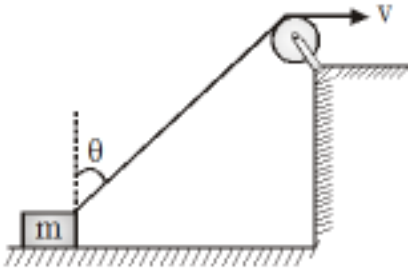
(1) $\left(3\hat{i} + 4\hat{j} + 12\hat{k}\right)$ kg

(2) 13 kg

(3) 5 kg

(4) None of these

3) A block is dragged on a smooth plane with the help of a rope which moves with a velocity v as shown in figure. The horizontal velocity of the block is:

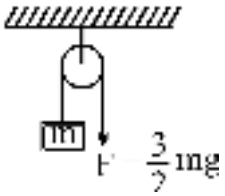


(1) v

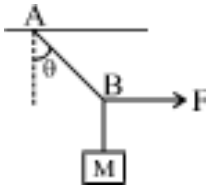
(2) $v \sin\theta$

(3) $\frac{v}{\sin\theta}$

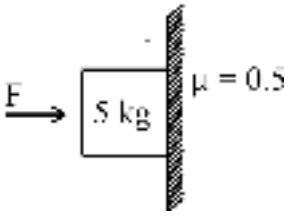
(4) $\frac{v}{\cos\theta}$

4) In the arrangement shown acceleration of the block is 

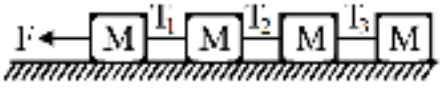
(1) Zero
(2) g/2
(3) g
(4) 2g

5) If the system is in equilibrium then tension in the string AB is 

(1) F sinθ
(2) F/sinθ
(3) F cos θ
(4) $\frac{F}{\cos\theta}$

6) Find out minimum value of F so that the 5 kg block remains at rest 

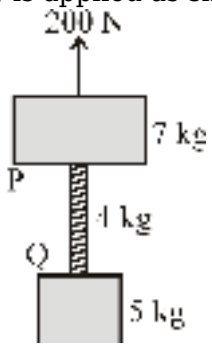
(1) 50 N
(2) 100 N
(3) 10 N
(4) 25 N

7) Four blocks of same mass connected by cords are pulled by a force F on a smooth horizontal surface, as shown in figure. The tensions T_1, T_2 and T_3 will be :- 

(1) $T_1 = \frac{1}{4}F, T_2 = \frac{3}{2}F, T_3 = \frac{1}{4}F$
(2) $T_1 = \frac{1}{4}F, T_2 = \frac{1}{2}F, T_3 = \frac{1}{2}F$
(3) $T_1 = \frac{3}{4}F, T_2 = \frac{1}{2}F, T_3 = \frac{1}{4}F$
(4) $T_1 = \frac{3}{4}F, T_2 = \frac{1}{2}F, T_3 = \frac{1}{2}F$

8) A block of 7 kg is connected with a block of 5 kg by a heavy rope of mass 4 kg. If an upward force

of 200 N is applied as shown in figure then the tension at the top of the rope at point P will be– (g = 10 m/s^2)



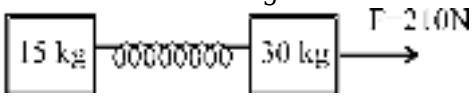
(1) 2.27 N

(2) 112.5 N

(3) 87.5 N

(4) 360 N

9) At the instant when the 15 kg mass has acceleration of 6 m/sec^2 then acceleration of 30 kg mass will be :



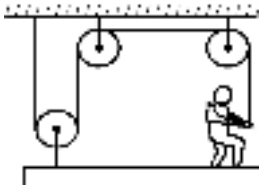
(1) 2 m/sec^2

(2) 3 m/sec^2

(3) 4 m/sec^2

(4) 6 m/sec^2

10) A 50 kg person stand on a 25 kg platform. He pulls on the rope which is attached to the platform via the frictionless pulleys as shown in the figure. The platform moves upwards at a steady rate if the force with which the person pulls the rope is



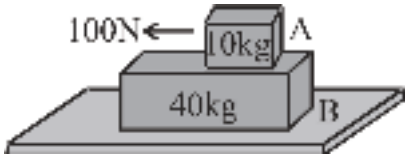
(1) 500 N

(2) 250 N

(3) 25 N

(4) None

11) A 40 kg slab rests on a frictionless floor as shown in the figure. A 10 kg block rests on the top of the slab. The static coefficient of friction between the block and slab is 0.60 while the kinetic friction is 0.40. The 10 kg block is acted upon by a horizontal force 100 N. If g = 10 m/s^2, the resulting acceleration of the slab will be :-



(1) 1 m/s^2

(2) 1.5 m/s^2

(3) 2 m/s^2

(4) 6 m/s^2

12) Two blocks of masses 1 kg and 2 kg are placed in contact on a smooth horizontal surface and a horizontal force 3N is applied on block 1 kg and an block 2 kg respectively in fig. (i) and fig. (ii) .If normal force of interaction between both blocks are N_1 and N_2 according to both fig (i) and fig (ii) respectively.



fig. (i) (N_1)

fig. (ii) (N_2)

Then the value of $\frac{N_1}{N_2}$ is :-

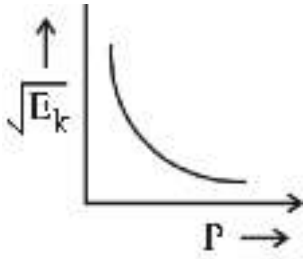
(1) $\frac{2}{1}$

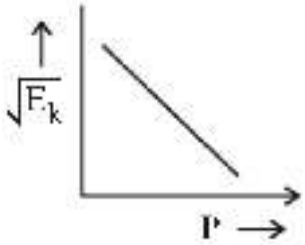
(2) $\frac{1}{2}$

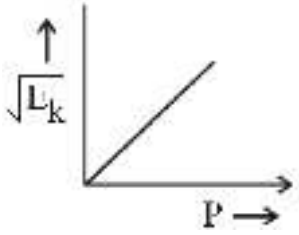
(3) $\frac{1}{4}$

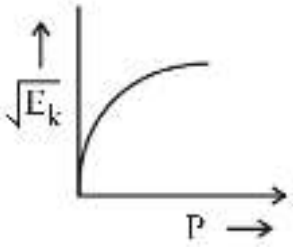
(4) $\frac{1}{3}$

13) The graph between $\sqrt{E_k}$ and p is -
(E_k = kinetic energy and P = momentum)

(1)


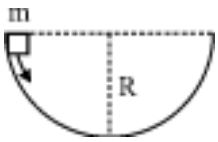
(2)


(3)


(4) 

14)

A mass m slips along the wall of a semispherical surface of radius R from rest. The velocity at the bottom of the surface is :-



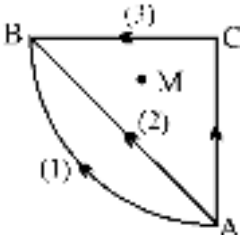
(1) $\sqrt{Rg}$

(2) $\sqrt{2Rg}$

(3) $2\sqrt{\pi Rg}$

(4) $0\sqrt{\pi Rg}$

15) In a region of only gravitational field of mass 'M' a particle is shifted from A to B via three different paths in the figure. The work done in different paths are W_1, W_2, W_3 respectively then



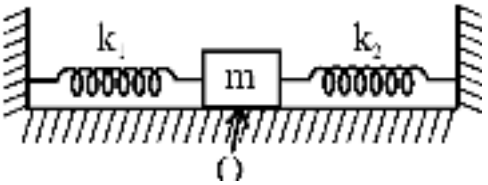
(1) $W_1 = W_2 = W_3$

(2) $W_1 = W_2 > W_3$

(3) $W_1 > W_2 > W_3$

(4) $W_1 < W_2 < W_3$

16) A block of mass m is attached to two unstreched springs of spring constants k_1 and k_2 as shown in figure. The block is displaced towards right through a distance x and is released. Find the speed of the block as it passes through the mean position O.



(1) $\sqrt{\frac{(k_1 + k_2)x^2}{m}}$

(2) $\sqrt{\frac{3}{4}\frac{(k_1 + k_2)x^2}{m}}$

(3) $\sqrt{\frac{(k_1 + k_2)x^2}{4m}}$

(4) $\sqrt{\frac{(k_1 + k_2)x^2}{2m}}$

17) A drop of rain water of mass 2 g falling from a height of 0.5 km hits the ground with speed 30 m/s. Then work done by resistive force of air is

(1) –9.10 J

(2) –8.75 J

(3) +0.10 J

(4) –8.25 J

18) A body of mass m starting from rest from origin moves along x-axis with constant power (P). Then relation between velocity and time is :-

(1) $v \propto t^{1/2}$

(2) $v \propto t^{2}$

(3) $v \propto t^{3}$

(4) $v \propto t^{3/2}$

19) The power of a water pump is 2kW. If g = 10 ms^{-2}, the amount of water it can raise in one minute to a height of 10 m is :-

(1) 2000 litre

(2) 1000 litre

(3) 100 litre

(4) 1200 litre

20) If the momentum of a body increases by 100% the percentage increase in its kinetic energy is :-

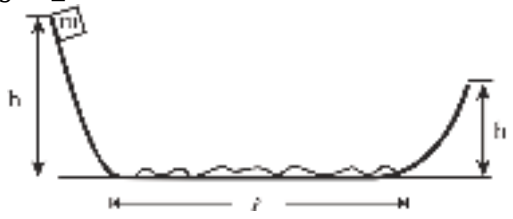
(1) 50%

(2) 100%

(3) 300%

(4) 500 %

21) A block is released from rest from a height h = 5m. After travelling through the smooth curved surface it moves on the rough horizontal surface through a length □ = 8 m and climbs onto the other smooth curved surface through a height h'. If μ = 0.5, find h' :-



(1) 2 m

(2) 3 m

(3) 1 m

(4) zero

22) A force of 1N newton is applied over a particle which displaces it from x = 0 to x = 7m. The work done on the particle is :-

(1) – 7 joules

(2) + 13 joules

(3) + 7 joules

(4) + 11 joules

23) A circular road of radius 1000 m has banking angle 45°. The maximum safe speed of a car having a mass 2000 kg will be, if the coefficient of friction between tyre and road is 0.5 :-

(1) 172 m/s

(2) 124 m/s

(3) 99 m/s

(4) 86 m/s

24) A stone of mass 1 kg tied to a light inextensible string of length L = $\frac{10}{3}$ m is whirling in a circular path of radius L in a vertical plane. If the ratio of the maximum tension in the string to the minimum tension in the string is 4 and if g is taken to be 10 m/sec^2, the speed of the stone at the highest point of the circle is

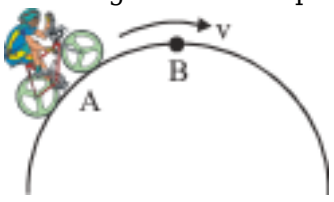
(1) 20 m/sec

(2) $10\sqrt{3}$ m/sec

(3) $5\sqrt{2}$ m/sec

(4) 10 m / sec

25) A motorcyclist is going on an overbridge of radius r maintaining a constant speed v. As the motorcyclist goes up on the overbridge the normal force :-



(1) Increases ($N_B > N_A$)

(2) Decreases ($N_B < N_A$)

(3) Remains same ($N_B = N_A$)

(4) Nothing can be said

26) A stone is tied to a string of length L is whirled in a vertical circle with the other end of the string at the centre. At a certain instant of time, the stone is at its lowest position and has a speed u. The magnitude of the change in its velocity as it reaches a position where the string is horizontal is :-

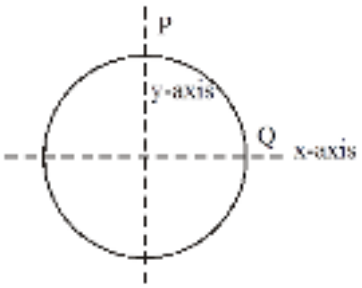
(1) $\sqrt{u^2 - 2gL}$

(2) $\sqrt{2gL}$

(3) $\sqrt{u^2 - gL}$

(4) $\sqrt{2(u^2 - gL)}$

27) A particle moves in a circle of radius 4 cm clockwise at constant speed 2 cm/s. The acceleration



of particle at instance half way between P & Q is :-

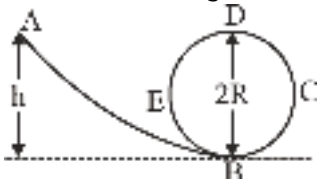
(1) $-4(\hat{i} + \hat{j})$

(2) $4(\hat{i} + \hat{j})$

(3) $-\left(\frac{\hat{i} + \hat{j}}{\sqrt{2}}\right)$

(4) $\left(\frac{\hat{i} - \hat{j}}{4}\right)$

28) A frictionless track ABCDE ends in a circular loop of radius R. A body slides down the track from point A which is at a height h = 5 cm. Maximum value of R for the body to successfully complete the



loop is :-

(1) 5 cm

(2) $\frac{15}{4}$ cm

(3) $\frac{10}{3}$ cm

(4) 2 cm

29)

If a_r and a_t represents radial and tangential acceleration, the motion of a particle will be circular if :-
(a) $a_r = 0$ and $a_t = 0$ (b) $a_r = 0$ but $a_t \neq 0$
(c) $a_r \neq 0$ but $a_t = 0$ (d) $a_r \neq 0$ and $a_t \neq 0$

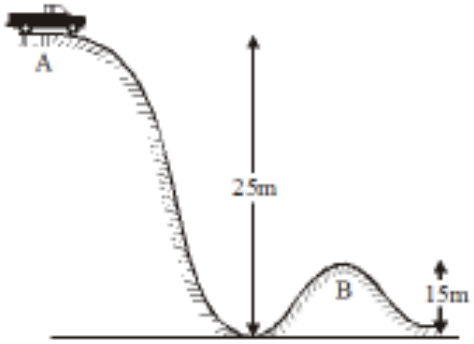
(1) c

(2) d

(3) a,b

(4) c,d

30) Figure shows the roller coaster track. Each car will start from rest at point A and will roll with negligible friction. It is important that there should be at least some small positive normal force exerted by the track on the car at all points, otherwise the car would leave the track. With the above fact, the minimum safe value for the radius of curvature at point B is ($g = 10\ m/s^2$) :



(1) 20 m

(2) 10 m

(3) 40 m

(4) 25 m

31) A coin is placed on the horizontal surface of a rotating disc. The distance of the coin from the axis is 1 m and coefficient of friction is 0.5. If the disc starts from rest and is given an angular acceleration 1 rad/s^2, the number of revolutions through which the disc turns before the coin slips is

(1) $\frac{4}{\pi}$

(2) $\frac{7}{4\pi}$

(3) $\frac{5}{4\pi}$

(4) $\frac{3}{4\pi}$

32) If the road is unbanked and the coefficient of friction between the road and the tyres is 0.8, then the maximum speed with which an automobile can move around a curve of 84.5 m radius without slipping is : (Take $g = 10\ ms^{-2}$) :-

(1) $26\ ms^{-1}$

(2) $67.7\ ms^{-1}$

(3) $13\ ms^{-1}$

(4) $36.7\ ms^{-1}$

33) Three identical particles are joined together by a thread as shown in figure. All the three particles are moving in a horizental plane, If the velocity of the outermost particle is v0.then the ratio of the tensions in the three section of the string is :



(1) 3 : 5 : 7

(2) 3 : 4 : 5

(3) 7 : 11 : 6

(4) 3 : 5 : 6

34)

A stone tied to the end of a 80 cm long string is whirled in a horizontal circle with a constant speed. If the stone makes 14 revolutions in 25 s, the magnitude of its acceleration is:

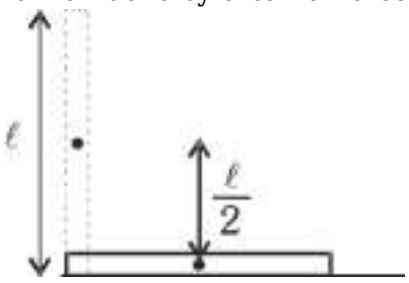
(1) 20 m/s^2

(2) 12 m/s^2

(3) 9.9 m/s^2

(4) 8 m/s^2

35) A rod of mass 'm' and length '□' is placed in horizontal position. Find workdone by external force against gravity to take it from horizontal position to vertical position.



(1) mg□

(2) mg□/2

(3) mg□/3

(4) zero

36) A bomb initially at rest explodes by it self into three equal mass fragments. The velocities of two fragments are $(3\hat{i} + 2\hat{j})$ m/s and $(-\hat{i} - 4\hat{j})$ m/s. The velocity of the third fragment (in m/s) is :-

(1) $2\hat{i} + 2\hat{j}$

(2) $2\hat{i} - 2\hat{j}$

(3) $-2\hat{i} + 2\hat{j}$

(4) $-2\hat{i} - 2\hat{j}$

37) Two ice skaters A and B approach each other at right angles. Skater A has a mass 30 kg and velocity 1 m/s and skater B has a mass 20 kg and velocity 2 m/s. They meet and cling together. Their final velocity is

(1) 2 m/s

(2) 1.5 m/s

(3) 1 m/s

(4) 2.5 m/s

38) A projectile of mass m is fired with velocity V from a point P, as shown, the magnitude of the change in momentum between the points P and arriving at Q is : 

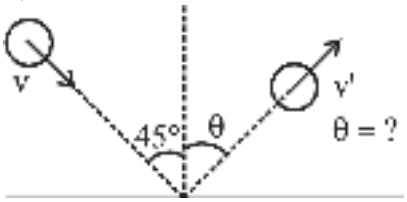
(1) 0

(2) $\frac{mV}{2}$

(3) $mV\sqrt{2}$

(4) 2 mV

39) A ball collides to a surface as shown in figure. If coefficient of restitution $e = \frac{1}{\sqrt{3}}$, then θ will be:-

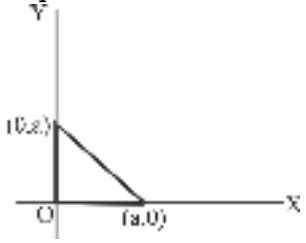


(1) 45°

(2) 60°

(3) 30°

(4) 0°

40) Three rods of the same mass are placed as shown in the figure. What will be the co-ordinates of the centre of mass of the system ? 

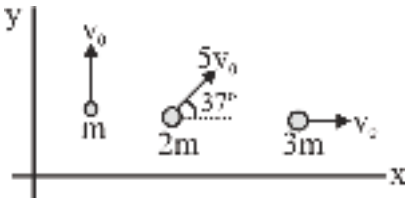
(1) $\left(\frac{a}{2}, \frac{a}{2}\right)$

(2) $\left(\frac{a}{\sqrt{2}}, \frac{a}{\sqrt{2}}\right)$

(3) $\left(\frac{\sqrt{2}a}{3}, \frac{\sqrt{2}a}{3}\right)$

(4) $\left(\frac{a}{3}, \frac{a}{3}\right)$

41) Find the velocity of centre of mass of given system :-



(1) $\frac{3v_0}{5}\hat{i} + \frac{7v_0}{5}\hat{j}$

(2) $2v_0\hat{i} + 3v_0\hat{j}$

(3) $v_0\hat{i} + v_0\hat{j}$

(4) $\frac{11v_0}{6}\hat{i} + \frac{7v_0}{6}\hat{j}$

42) A bullet of mass 10g moving horizontally with a velocity of 400 m/s strikes a wooden block of mass 2 kg which is suspended by a light inextensible string of length 5 m. As a result, the centre of gravity of the block is found to rise a vertical distance of 10 cm. The speed of the bullet after it emerges out horizontally from the block will be :-

(1) 120 m/s

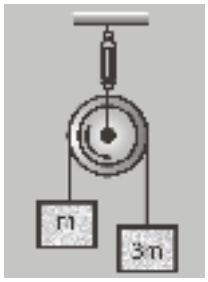
(2) 160 m/s

(3) 100 m/s

(4) 80 m/s

43)

If the system is released, then the acceleration of the centre of mass of the system :-



(1) g/4

(2) g/2

(3) g

(4) 2g

44) **Assertion :** The centre of mass of a system of two particles is closer to the heavier particle.
Reason : The centre of mass can be outside the body also.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true and Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

45) In an inelastic collision between two bodies, the physical quantity that is conserved :-

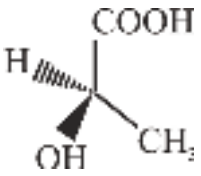
(1) Kinetic energy

(2) Momentum

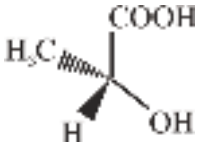
(3) Potential energy

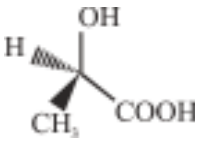
(4) Kinetic energy and momentum

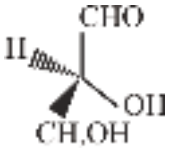
CHEMISTRY



1) Which of the following compound is enantiomeric with given compound ?

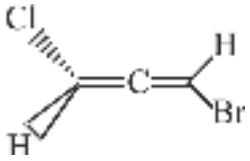
(1) 

(2) 

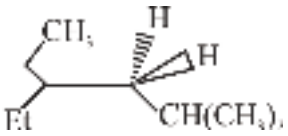
(3) 

(4) None of the above

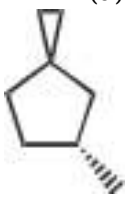
2) Which of the following set molecules is (are) chiral?



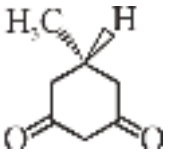
(a)



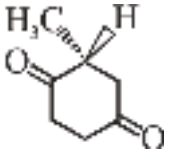
(b)



(c)



(d)



(e)

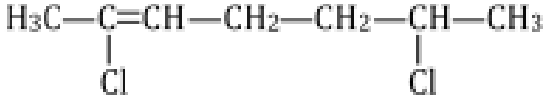
(1) a, d

(2) a, b

(3) a, c

(4) d, e

3)



Total stereoisomers of compound 'x' is :

(1) 2

(2) 8

(3) 6

(4) 4

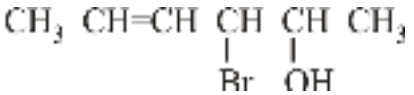
4) Select the chiral molecules. (a) d-Tartaric acid (b) Lactic acid
(c) Acetic acid (d) Adipic acid

(1) a, b and c

(2) a and d

(3) a and b

(4) a, b, c and d

5) Number of stereo isomer of the given compound is : 

(1) 2

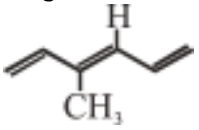
(2) 4

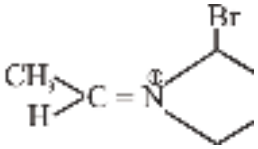
(3) 6

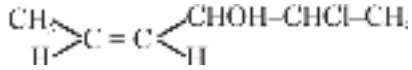
(4) 8

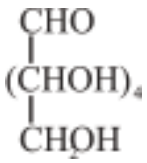
6)

Arrange the following compounds in decreasing order for stereogenic centres :-

(I) 

(II) 

(III) 

(IV) 

Select the correct answer from the codes given below :

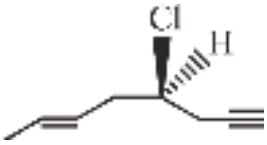
Codes :-

(1) I > II > III > IV

(2) IV > III > II > I

(3) IV > III > I > II

(4) I > III > IV > II

7)  is :-

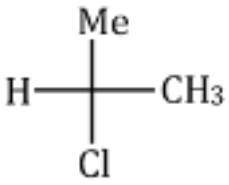
(1) (4R, 6E)-4-chloro oct-6-en-1-yne

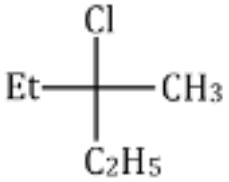
(2) (4S, 6E)-4-chloro oct-6-en-1-yne

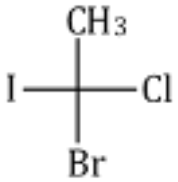
(3) (5R, 2E)-5-chloro oct-2-en-7-yne

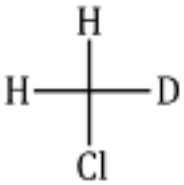
(4) (5S, 2E)-5-chloro oct-2-en-7yne

8) Which of the following compounds has chiral carbon atom?

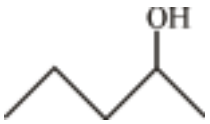
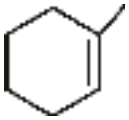
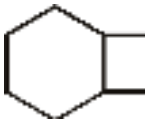
(1) 

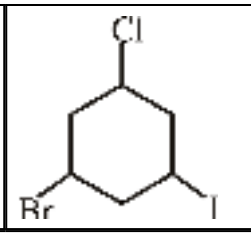
(2) 

(3) 

(4) 

9) Match the following:-

Compound		Chiral-C-atom	
(A)		(P)	3
(B)		(Q)	2
(C)		(R)	0

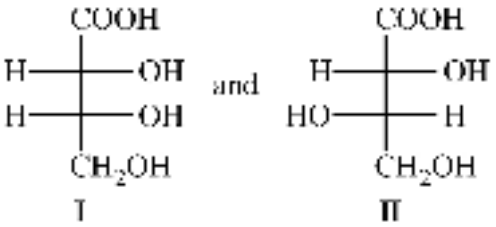
(D)		(S)	1

(1) A – P, B – Q, C – R, D – S

(2) A – S, B – R, C – Q, D – P

(3) A – S, B – R, C – P, D – Q

(4) A – S, B – P, C – R, D – Q



10) The correct statement regarding compared I and II is :-

(1) I is meso & II is active

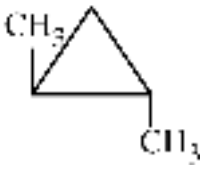
(2) II is meso and I is active

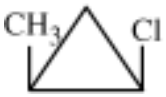
(3) Both I and II are active

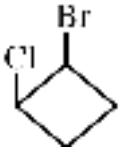
(4) Both I and II are meso

11) In which of the following compounds, plane of symmetry is present ?

(1) 

(2) 

(3) 

(4) 

12) Consider the following statements

a. If a compound rotates plane polarized light in clockwise direction then it is called dextrorotatory

b. A molecule which is non-superimposable on its mirror image is chiral and optically active

c. Propan-2-ol is an achiral molecule

The correct statements are

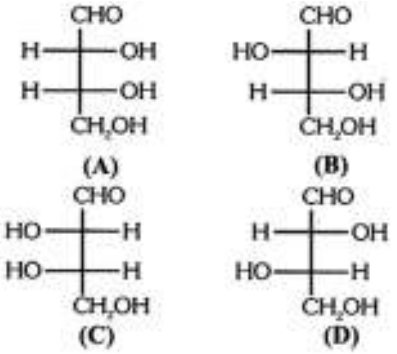
(1) a and b only

(2) b and c only

(3) a, b and c

(4) a and c only

13) The correct corresponding order names of four aldoses with configuration given below



(1) L-erthyrose, L-threose, D-erythrose, D-threose

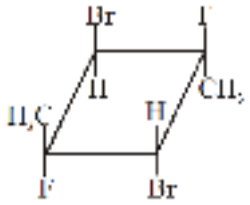
(2) D-erythrose, D-threose, L-erthrose, L-threose

(3) L-erythrose, L-threose, L-erythrose, D-threose

(4) D-threose, D-erythrose, L-threose, L-erythrose

14)

The compound has :-



(1) Plane of symmetry

(2) Center of symmetry

(3) Two pos

(4) No symmetry

15)

The simplest alcohol that can exhibit enantiomerism :-

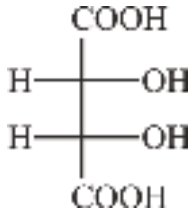
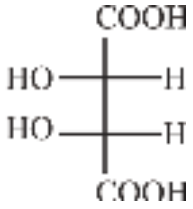
(1) 1-propanol

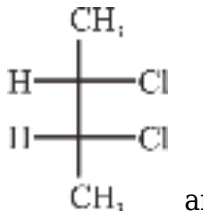
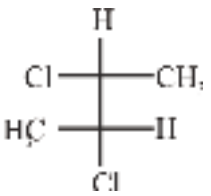
(2) 2-propanol

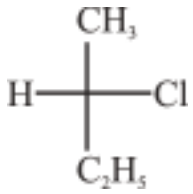
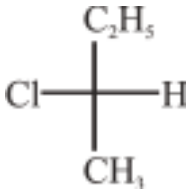
(3) 2-butanol

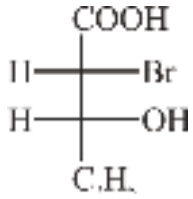
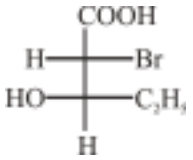
(4) 1-butanol

16) Which of the following is correct statement:-

(1)  and  : Enantiomer

(2)  and  : Diastereomer

(3)  and  : Enantiomers

(4)  and  : Identical (Homomer)

17) Correct order of pK_a is

(i) CH_3-CH_2-OH (ii) $H_3C-\underset{\displaystyle CH_3}{\underset{|}{CH}}-OH$ (iii) $H_3C-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-OH$

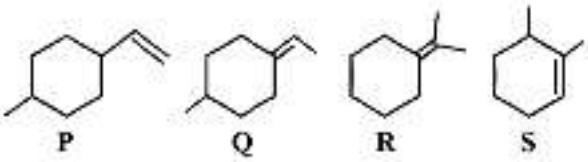
(1) i > ii > iii

(2) ii > i > iii

(3) iii > ii > i

(4) ii > iii > i

18) Arrange the following alkenes in increasing order of their heat of hydrogenation :-



(1) R < Q < S < P

(2) R < S < P < Q

(3) P < Q < R < S

(4) P < Q < S < R

19) **Assertion :** Heterolytic fission occurs readily in polar covalent bonds.

Reason : Heterolytic fission involves breaking of bond in such a way that the shared pair of electrons go with one atom.

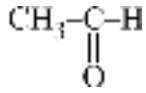
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion

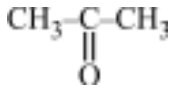
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion

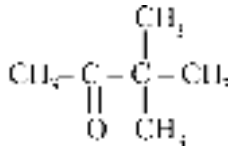
(3) Assertion is True but the Reason is False

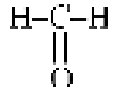
(4) Both Assertion & Reason are False

20) Which of the following does not shows tautomerism ?

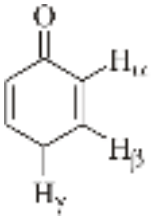
(1) 

(2) 

(3) 

(4) 

21) In the enolization of the given molecule, the H-atom involved is



(1) α-H

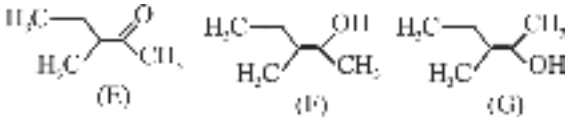
(2) β-H

(3) γ-H

(4) cannot be enolized

22)

Which statement is incorrect for the structures E, F and G :-



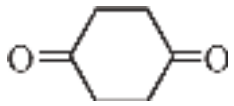
(1) E, F and E, G are tautomers

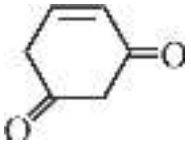
(2) F and G are geometrical isomers

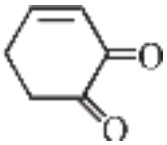
(3) E, F and G are resonating structures

(4) F and G are diastereomers

23) Which of the following does not show tautomerism :-

(1) 

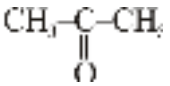
(2) 

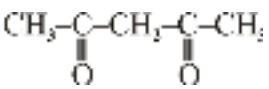
(3) 

(4) 

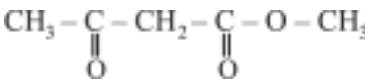
24)

Which has most acidic H :

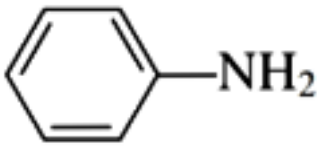
(1) 

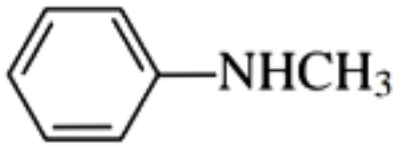
(2) 

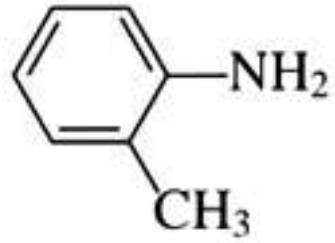
(3) 

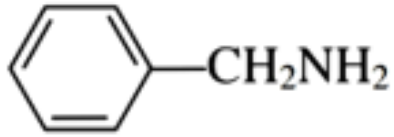
(4) 

25) Which of the following is the strongest base -

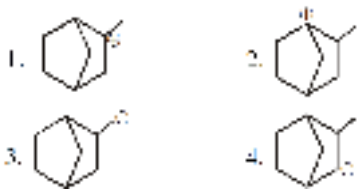
(1) 

(2) 

(3) 

(4) 

26) Find out correct stability order in the following carbocations :-



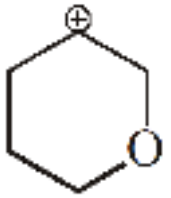
(1) 4 > 1 > 3 > 2

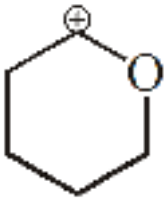
(2) 1 > 4 > 3 > 2

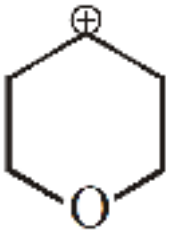
(3) 1 > 2 > 4 > 3

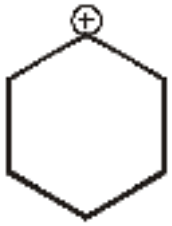
(4) 2 > 1 > 4 > 3

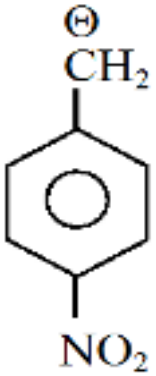
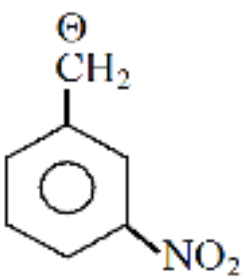
27) Most stable carbocation is :-

(1) 

(2) 

(3) 

(4) 

28)  is more stable than  , because

(X) (Y)

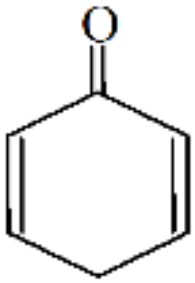
(1) 180º location of NO2 and $\overset{\ominus}{C}H_2$ in X

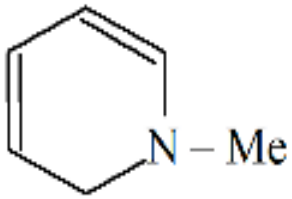
(2) – NO_2 opperate both –I and – M in X.

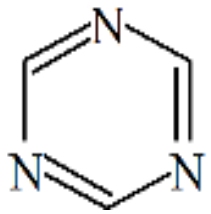
(3) – I in Y is weaker than X

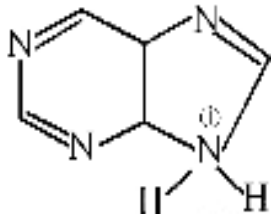
(4) Due to steric repulsion

29) Which of following compound is aromatic ?

(1) 

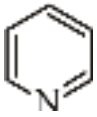
(2) 

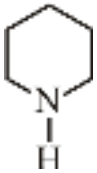
(3) 

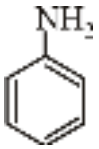
(4) 

30) Most basic compound is

(1) 

(2) 

(3) 

(4) 

31) Which set of anions gives same colour gas with dilute acid and concentrated acid respectively.

(1) Br^-, NO_3^-

(2) NO_2^-, NO_3^-

(3) I^-, NO_2^-

(4) NO_2^-, Br^-

32) Dimethylglyoxime is used for the test of ..... in alkaline medium

(1) Cobalt

(2) Zinc

(3) Manganese

(4) Nickel

33) When a mixture of solid NaCl, solid $K_2Cr_2O_7$ is heated with concentrated H_2SO_4, orange red vapours are obtained.These are of the compound :-

(1) chromous chloride

(2) chromyl chloride

(3) chromic chloride

(4) chromic sulphate

34) During salt analysis basic radicals, precipitate formation will takes place, when:

(1) $K_{ip} < K_{sp}$

(2) $K_{ip} > K_{sp}$

(3) Never

(4) Always

35) A mixture, on heating with concentration H_2SO_4 and MnO_2, librates reddish brown vapour of

(1) Br_2

(2) NO_2

(3) HBr

(4) I_2

36) At the occasion of marriage, the fire works are used, which of the following gives green flame ?

(1) Ba

(2) K

(3) Be

(4) Na

37) During the testing of basic radicals the group reagent for group II is:

(1) dil HCl

(2) NH_4OH in presence of NH_4Cl

(3) H_2S in presence of HCl

(4) H_2S in presence of NH_4OH and NH_4Cl

38) Which of the following will give chocolate brown precipitate on adding potassium ferrocyanide ($K_4[Fe(CN)_6]$) ?

(1) $ZnCl_2$

(2) $FeCl_3$

(3) $CuSO_4$

(4) AgCl

39) An aqueous solution of given salt, when treated with dil. HCl, gives colourless gas with smell like rotten eggs, is indicator of presence of :

(1) CO_3^{2-}

(2) S^{2-}

(3) NO_2^-

(4) All

40) Incorrect match is :

	Identification test	**Colour**
(1)	Starch - iodine test	Blue
(2)	Chromyl chloride test	Green
(3)	Layer test of bromide	Red-Brown
(4)	Layer test of Iodide	Violet

(1) 1

(2) 2

(3) 3

(4) 4

41)

	List-I (Basic Radical)		**List-II (Group)**
(P)	Pb^{+2}	(1)	VI group

(Q)	Fe^{+3}	(2)	IV group
(R)	Ni^{+2}	(3)	I group
(S)	Mg^{+2}	(4)	III group

Correct matching is :

	P	**Q**	**R**	**S**
(1)	1	3	2	4
(2)	3	4	2	1
(3)	4	3	2	1
(4)	3	4	1	2

(1) 1

(2) 2

(3) 3

(4) 4

42) Ca and Ba ions are precipitated in fifth group as their :

(1) O^{-2}

(2) SO_4^{-2}

(3) CO_3^{-2}

(4) S^{-2}

43) On heating a small lump of borax, a glassy bead is obtained which contain :-

(1) $Na_2B_4O_7$

(2) $NaBO_2$ & B_2O_3

(3) Na_2O

(4) Na_2B

44)

	List-I (Molecule)		**List-II (Characteristic odour)**
(A)	CO_2	(P)	Rotten egg smell
(B)	I_2	(Q)	Violet Fumes
(C)	H_2S	(R)	Reddish brown vapours
(D)	Br_2	(S)	Odour less

(1) A→S, B→Q, C→P, D→R

(2) A→S, B→P, C→Q, D→R

(3) A→S, B→R, C→Q, D→P

(4) A→P, B→Q, C→R, D→S

45)

Match **List-I** with **List-II** :

	List-I (Test/reagent)		**List-II (Radical identified)**
A.	Lake Test	I.	NO_3^-
B.	Nessler's Reagent	II.	Fe^{3+}
C.	Potassium sulphocyanide thiocyanate	III.	Al^{3+}
D.	Brown Ring Test	IV.	NH_4^+

Choose the **correct** answer from the options given below :

(1) A-IV, B-II, C-III, D-I

(2) A-II, B-IV, C-III, D-I

(3) A-II, B-III, C-IV, D-I

(4) A-III, B-IV, C-II, D-I

BIOLOGY

1) Major regulatory enzyme of glycolysis –

(1) Phosphofructokinase

(2) Hexokinase

(3) Aldolase (Lyases)

(4) Pyruvate decarboxylase

2) Which of the following is correct step?

(1) Pyruvic acid + NAD^+ $\xrightarrow[\text{dehydrogenase}]{\text{Pyruvate}}$ Acetyl-CoA + NADH + H^+

(2) Pyruvic acid + NAD^- $\xrightarrow[\text{Phosphodikinase}]{Mg^{++}}$ Acetyl-CoA + NAD^+

(3) Pyruvic acid $\xrightarrow[\text{Phosphodikinase}]{Mg^{+2}}$ Acetyl-CoA + NADH

(4) Pyruvic acid + CoA + NAD^+ $\xrightarrow[\text{Pyruvate dehydrogenase}]{Mg^{+2}}$ Acetyl-CoA + NADH + H^+ + CO_2

3) Complete the following reaction-

$$\text{Pyruvic acid} + \text{(i)} + \text{NAD}^{+} \xrightarrow[\text{Enzyme}]{\text{(ii)}} \text{Acetyl COA} + \text{(iii)} + \text{NAD} + \text{H}^{+}$$

(1) (i) O_2 (ii) Mg^{2+} (iii) CO_2

(2) (i) O_2 (ii) Na^{+} (iii) H_2O

(3) (i) CoA (ii) Na^{+} (iii) CO_2

(4) (i) CoA (ii) Mg^{2+} (iii) CO_2

4) The Respiratory Quotient (RQ) value for glucose is _____.

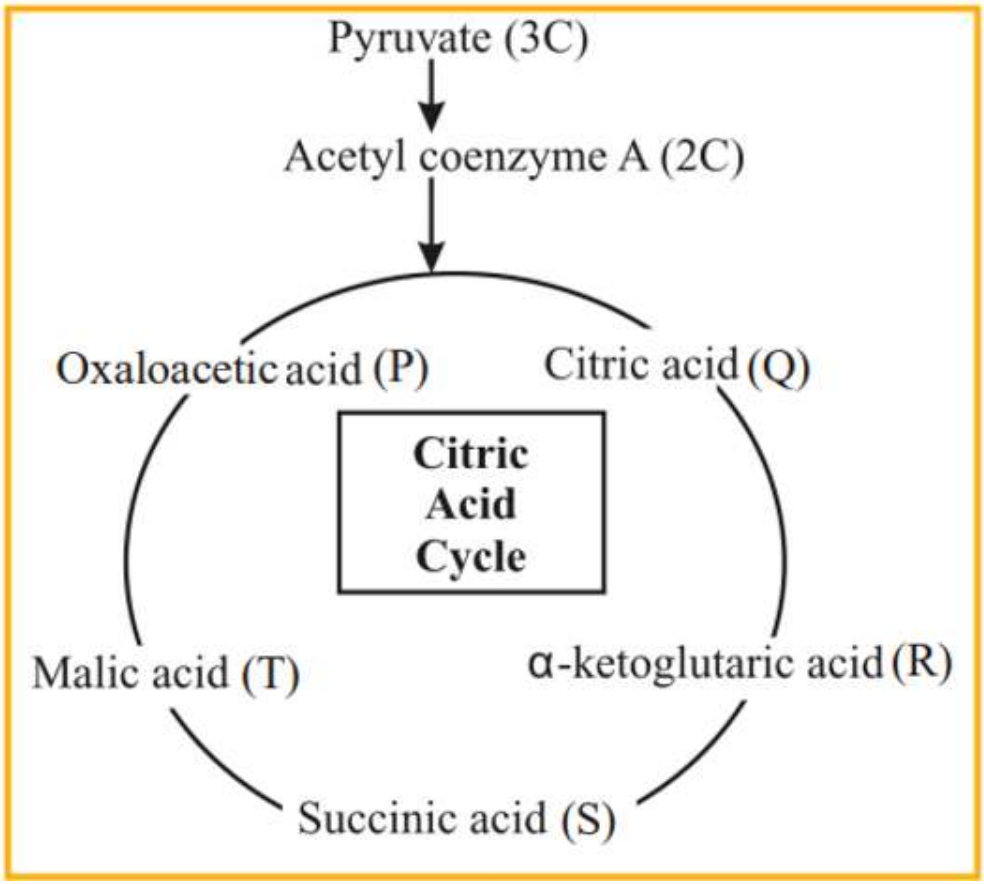
(1) Less than 1

(2) Greater than 1

(3) Equal to 1

(4) Zero

5) Refer the figure of citric acid cycle and choose the correct combination of labelling (P, Q, R, S) the number of carbon compounds in the substrate molecules, involved in the given figure.



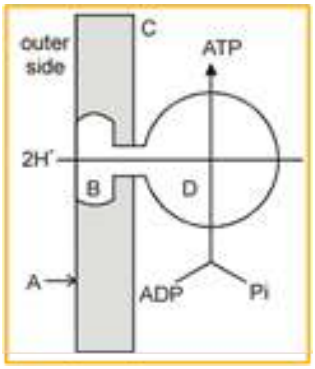
(1) (P) 4C, (Q) 6C, (R) 5C, (S) 4C

(2) (P) 6C, (Q) 5C, (R) 4C, (S) 3C

(3) (P) 2C, (Q) 5C, (R) 6C, (S) 4C

(4) (P) 4C, (Q) 6C, (R) 4C, (S) 4C

6) Identify A, B, C and D in the following diagram –



(1) A - Inner membrane of chloroplast; B - F_0; C - F_1; D - Matrix

(2) A - Inner mitochondrial membrane; B - F_0; C - Matrix; D - F_1

(3) A - Matrix; B - F1; C - Inner mitochondrial membrane; D - F_0

(4) A - Cytoplasm; B - F_0; C - Matrix; D - F_1

7) The correct sequence of electron transporters in ATP synthesis is-

(1) cyt a a b c1

(2) cyt b c1 c a a3

(3) cyt b c1 c a3 a

(4) cyt c b a a3

8) Which enzyme catalyzes the initial step of the Tricarboxylic Acid cycle where acetyl group condenses with oxaloacetic acid to form citric acid?

(1) Citrate synthase

(2) Citrate isomerase

(3) Aconitase

(4) Succinate dehydrogenase

9) Find odd one out w.r.t. product of respiration

(1) O_2

(2) CO_2

(3) H_2O

(4) Energy

10) Which one is correct for glycolysis :-

(1) 2, O_2 used up

(2) 2, CO_2 - released

(3) 2, Pyruvic acid produced

(4) 2, $NADPH_2$ produced

11) Pyruvate may form :-

(1) Ethanol

(2) Lactic acid

(3) Acetyl-CoA

(4) Any of the above

12) Which is not true for glycolysis ?

(1) End product is CO_2 and H_2O

(2) Substrate level phosphorylation

(3) Production of ATP

(4) Expenditure of ATP

13) $2(C_{51}H_{98}O_6) + 145O_2 \rightarrow 102CO_2 + 98\ H_2O$ + Energy
RQ for the given reaction will be :-

(1) More than one

(2) Equal to RQ of carbohydrate

(3) Less than RQ of proteins

(4) More than RQ of organic acid

14) A - Occur in the cytoplasm of the cell.
B - Process is independent of oxygen.
C - It can occur in both aerobic and anaerobic conditions.
Choose the correct option for given statements A to C :-

(1) Krebs cycle

(2) Link reaction

(3) Glycolysis

(4) Fermentation

15) During link reaction _____ molecules of NADH are produced from the metabolism of _____ molecule of _____. Choose the correct answer for blanks :-

(1) two, two, Acetyl Co-A

(2) two, one, Glucose

(3) two, one, Pyruvic acid

(4) one, two, glucose

16) Fatty acid for entry into kreb's cycle degraded to:-

(1) Pyruvate

(2) Citric acid

(3) Acetyl CoA

(4) PGA

17)

(A) NADH is oxidised to NAD^+ for continuity of fermentation process.

(B) Pure proteins and fats are never used as respiratory substrate

(1) Both (A) and (B) are correct

(2) Both (A) and (B) are incorrect

(3) Only (A) is correct

(4) Only (B) is correct

18) How many molecules of ATP are directly synthesised in TCA cycle by oxidation of one glucose molecule ?

(1) 4

(2) 2

(3) 6

(4) 8

19) Pyruvic acid converted into ethyl alcohol by -

(1) Pyruvate decarboxylase

(2) Alcohol dehydrogenase

(3) Hexokinase

(4) Both (1) & (2)

20) FAD is an electron acceptor during oxidation of which of the following ?

(1) α-ketoglutaric acid → succinyl Co-A

(2) Succinic acid → Fumaric acid

(3) Succinyl CoA → Succinic acid

(4) Fumaric acid → Malic acid

21) Which of the following is incorrect about fermentation ?

(1) It account for partial breakdown of glucose

(2) Net gain is only 2 ATP

(3) NADH to NAD^+ formation reaction is vigorous

(4) It occur in cytoplasm

22) The passing of the electrons removed as part of the hydrogen atoms to molecular oxygen with synthesis of ATP, the site of this process located:-

(1) In the matrix of the mitochondria

(2) In the cytoplasm

(3) In the intermembrane space of mitochondria

(4) On the inner membrane of the mitochondria

23) Which of the following is not a function of water in plants?

(1) It is responsible for cell elongation
(2) It maintains the turgidity of cells
(3) It acts as a source of energy
(4) It provides medium for enzymatic activities

24) Heterophylly means

(1) The appearance of different forms of leaves on the same plant species
(2) The appearance of same form of leaves on different plants
(3) The appearance of same form of flowers on different plants
(4) The appearance of different forms of fruits on the same plant

25) Plant growth regulators (PGRs) are

(1) Simple organic substances of different chemical composition
(2) Complex organic substances of different chemical composition
(3) Simple and complex organic substances of same chemical composition
(4) Small organic substances of same chemical composition

26) Two synthetic auxins are

(1) IAA and IBA
(2) 2, 4-D and NAA
(3) IAA and NAA
(4) 2, 4-D and IBA

27) Which of the following is not a function of gibberellic acid?

(1) Bolting
(2) Delayed senescence
(3) Seed germination
(4) Ripening

28) Cytokinins are said to be antiageing hormone because they delay the senescence by

(1) Controlling mobilisation of nutrients
(2) Promoting catabolic reactions
(3) Decreased morphogenesis and high respiration
(4) Both (1) and (2)

29) The phenomenon where under certain conditions differentiated cells regain their ability to divide is termed as

(1) Development
(2) Plasticity
(3) Dedifferentiation

(4) Redifferentiation

30) The 'bakane' (foolish seedling) disease of rice seedlings, was caused by a fungal pathogen …A….. and led to the discovery of …B… . Select the correct option to fill in the blanks A and B

(1) A–*Gibberella fujikuroi,* B–Gibberellins
(2) A–*Trichoderma,* B–Auxins
(3) A–*Alternaria*, B–Cytokinins
(4) A–*Aspergillus,* B–Kinetin

31) Which of the following PGRs act antagonistically w.r.t. apical dominance?

(1) Auxins and Ethylene
(2) Auxins and Cytokinins
(3) ABA and GA
(4) Ethylene and ABA

32) In TCA cycle, how many reduced cofactors are produced from one Acetyl CoA?

(1) 3 NADH2, 1 FADH2
(2) 2 NADH2, 1 FADH2
(3) 4 NADH2, 2 FADH2
(4) 5 NADH2, 1 FADH2

33) Number of total ATP generated in TCA cycle per Acetyl CoA molecule is

(1) 10
(2) 12
(3) 14
(4) 24

34) Match the following

	Column I		**Column II**
a.	Auxin	(i)	Root hair formation
b.	Cytokinin	(ii)	Seed development
c.	Ethylene	(iii)	Xylem differentiation
d.	ABA	(iv)	Nutrient mobilisation

(1) a(iv), b(ii), c(iii), d(i)
(2) a(ii), b(iii), c(i), d(iv)
(3) a(i), b(iii), c(ii), d(iv)
(4) a(iii), b(iv), c(i), d(ii)

35) Which one of the PGR would you use if you are asked to :-
(A) Induce growth in axillary bud
(B) 'Bolting' in rosette plants
(C) Induce immediate stomatal clossure in leaves

(1) A–Cytokinin, B–GA, C–ABA
(2) A–ABA, B–Ethylene, C–GA
(3) A–Auxin, B–Ethylene, C–GA
(4) A–Cytokinin, B–Ethylene C–ABA

36) Typical growth curve in plant is :-

(1) Linear
(2) Sigmoid
(3) Parabolic
(4) Both 1 & 2

37) In tissue culture low cytokinins to high auxin ratio cause :

(1) Root differentiation
(2) Shoot differentiation
(3) Both (1) and (2)
(4) Seed formation

38) The exponential growth can be mathematically expressed as :

(1) $L_t = L_0 + rt$
(2) $W_t = W_0 + e^{rt}$
(3) $W_t = W_0\, e^{rt}$
(4) $L_t = L_0 - rt$

39) 

Choose the correct option for A,B,C or D :-

(1) A– Root apical meristem
(2) B – Cork cambium
(3) C – Shoot apical meristem
(4) D – Primary growth

40) Plant growth regulator, which was first isolated from a fungus, is not related to which of the following physiological effects ?

(1) Increasing the yield in sugarcane crop

(2) Early seed production in conifers

(3) Root initiation in twigs

(4) Bolting in rosette habit plants

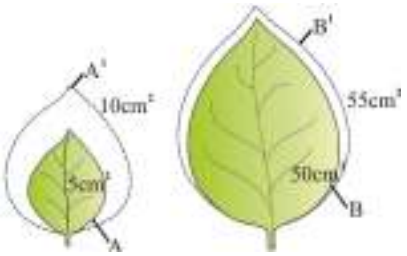
41) Charles Darwin and his son Francis Darwin found that coleoptile tip of a plant responded to unilateral illumination by growing towards the light source (phototropism). They worked with plant?

(1) Phalaris (Canary grass)

(2) Avena (Oat)

(3) Saccharum

(4) Triticale

42) 

Choose the incorrect statement with respect to leaf A & B (A' & B' as their final size respectively) given in the diagram.

(1) Relative growth rate of A is twice as that of B.

(2) Absolute growth of both A & B are same.

(3) Relative growth rate of B is less than A.

(4) Relative growth rate of B is 1/10 of A.

43) Exponential growth equation is '$W_1 = W_0 e^{rt}$'. In this equation 'r' is

(1) Relative growth rate and is also the measure of efficiency index of the plant

(2) Base of natural logarithms

(3) Time of growth

(4) Initial size

44)

Identify the correct match from column I, II and III

	Column I		**Column II**		**Column III**
1	GA	a	Gaseous	i	Promote bolting
2	Auxin	b	Cytokinesis	ii	Zeatin
3	Cytokinin	c	Phototropism	iii	Flowering in Pineapple

4	Ethylene	d	Acidic	iv	Fruit ripening

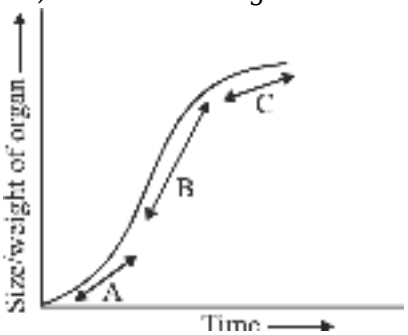
(1) 1–d–ii, 2–a–iv, 3–c–iii, 4–b–i

(2) 1–d–i, 2–c–iii, 3–b–ii, 4–a–iv

(3) 1–c–i, 2–b–iv, 3–a–ii, 4–d–iii

(4) 1–b–ii, 2–c–i, 3–d–iii, 4–b–iv

45) Observe the sigmoid curve given below and identify the phases A, B and C :-



(1) A = Initial slow growth, B = Rapid growth, C = Phase of growth during limited nutrient supply

(2) A = Rapid growth, B = Initial slow growth, C = Phase of growth during limited nutrient supply

(3) A = Lag phase, B = Stationary phase, C = Exponential phase

(4) A = Exponential phase, B = Stationary phase, C = Lag phase

46) Select the incorrect statement w.r.t. acromegaly.

(1) It is a disorder caused due to hyposecretion of hormone secreted from pituitary gland.

(2) It generally occurs in middle age person.

(3) It results in severe disfigurement of the face.

(4) It is hard to diagnose in the early stages and often goes undetected for many years

47) Read the following statements and choose the **correct** option.
statement A : Small amounts of androgenic steroids are secreted by the adrenal medulla which play a role in the growth of axial hair, pubic hair, and facial hair during puberty.
statement B : Cortisol is involved in maintaining the cardio vascular system as welI as the kidney functions.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

48) Consider the following features:
(a) Released by neurohypophysis
(b) Acts on the smooth muscles of uterus and stimulates their contractions during partuntion
(c) Acts on the mammary gland for the ejection of milk
The above given features are associated with

(1) Progesterone

(2) Oxytocin

(3) Estrogen
(4) Prolactm

49) The hormone androgens

(1) Are secreted from testis under the influence of FSH
(2) Influence the male sexual behaviour
(3) Produce anabolic effects on only protein metabolism
(4) Are responsible for high pitch of voice in males

50) Sella tursica is a depression in the sphenoid bone which lodges

(1) Pineal gland
(2) Pituitary gland
(3) Hypothalamus
(4) Thyroid gland

51) **Assertion (A):** Exopthalmic goitre is a disease characterised by enlargement of the parathyroid gland, increased BMR, weight loss, protrusion of eyeballs, etc.
Reason (R): Exophtalmic goitre is a form of hyperparathyroidism.
In the light of above statements, select the **correct** option.

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(2) Both (A) and (R) are true and (R) is the correct explanation of (A)
(3) Both (A) and (R) are false
(4) (A) is true but (R) is false

52) Due to malfunctioning of insulin hormone, what effect would be on blood sugar and disease occur :

(1) Hyperglycemia, Diabetes Mellitus
(2) Hypoglycemia, Diabetes Insipidus
(3) Hypoglycemia, Diabetes Mellitus
(4) Hyperglycemia, Diabetes Insipidus

53) Match the column-B with column-B :

	A		**B**
A	Parafollicular cells	(i)	Aldosteron
B	Parathyroid gland	(ii)	Calcitonin hormone
C	Thyroid gland	(iii)	Collips hormone
D	Zona glomerulosa	(iv)	Tetra iodothyronin

(1) A-iii, B-ii, C-iv, D-i
(2) A-ii, B-i, C-iii, D-iv

(3) A-ii, B-iii, C-iv, D-i

(4) A-iii, B-i, C-ii, D-iv

54) Match the gland, hormone and action :-

(1) Corpus luteum - Estrogen - Supports pregnancy

(2) Thyroid gland - Thyroxine - Regulate blood calcium level

(3) Anterior pituitary - oxytocin - Child birth

(4) Posterior pituitary - Vasopressin - Stimulates reabsorption of water from distal tubules.

55) Which group of hormones is responsible for the growth of axial hairs, pubic hairs and facial hairs during puberty ?

(1) Androgenic steroids

(2) Gluco-corticoids

(3) Mineralo-corticoids

(4) Both (2) and (3)

56) Which of the following statement is false :-

(1) Androgens play a stimulatory role in the process of spermatogenesis

(2) Androgens act on the central neural system and influence the male sexual behaviour

(3) Androgens stimulate high pitch of voice in male

(4) Androgens produce anabolic effect on protein metabolism

57) Read the following statements (A-D) :-
(A) Glucagon is a peptide hormone and plays an important role in maintaining the normal blood glucose levels.
(B) Glucagon acts mainly on the liver cells and stimulates glycogenolysis resulting in an increased blood sugar.
(C) Glucagon reduces the cellular glucose uptake and utilisation.
(D) Glucagon stimulates gluconeogenesis which contributes to hyperglycemia.
How many of the above statements are true :-

(1) 3

(2) 4

(3) 2

(4) 1

58) Which of the following is not the function of cortisol :-

(1) Stimulate formation of lipids in adipose tissue

(2) Stimulate proteolysis

(3) Anti-inflammatory action

(4) Immuno - suppressive action

59) Which of the following is not present in severe cases of diabetes mellitus :-

(1) Diabetic coma

(2) Ketoacidosis

(3) Hypoglycemia

(4) Ketonuria

60) Select the answer which correctly matches the endocrine gland with the hormone it secretes and its function/ deficiency symptom:
Endocrine gland Hormone Function/ deficiency/ symptoms

(1) Posterior pituitary Growth Hormone (GH) Oversecretion stimulates abnormal growth

(2) Thyroid gland Thyroxine Lack of iodine in diet results in goitre

(3) Corpus luteum Testosterone Stimulates spermatogenesis

(4) Anterior pituitary Oxytocin Stimulates uterus contraction during child birth

61) A : Our body secretes adrenaline in parasympathetic stimulation mainly.
R : Adrenaline decreases metabolic rate.

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is not the correct explanation of A.

(3) A is true but R is false.

(4) Both A and R are false.

62) A : Oxytocin stimulates dilation of uterine muscles during birth and prevent ejection of milk.
R : It is synthesized in the posterior lobe of pituitary.

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is not the correct explanation of A.

(3) A is true but R is false.

(4) Both A and R are false.

63) Assertion (A):– Both FSH and LH hormones are secreted from pars distalis, collectively called as gonadotrophins.
Reason (B):– These hormones acts on testis in males and ovaries in females and stimulate the gonadal ativity.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) Both (A) and (R) are false.

64) According to the accepted concept of hormone action if receptor molecules are removed from target organs.

(1) The target organ will continue to respond to hormone without any difference.

(2) The target organ will continue to respond to the hormone but will require higher concentration.

(3) The target organ will not respond to the hormone

(4) The target organ will continue to respond to the hormone but in a opposite way.

65) Consider the following four statement (A–D) and select option which include all the correct ones only :-
(A) Acromegaly is due to hyper secretion of STH in childhood stage.
(B) In adult female hypothyroidism cause irregular menses.
(C) Hyper secretion of glucocorticoids cause "Addison's disease".
(D) At the time of physical labour or fasting taking of insulin injection can cause ‘insulin shock’ in diabetic patient.

(1) Statements B & C
(2) Statements B & D
(3) Statements A & B
(4) Statements A & C

66) Read the following statement and select correct answer.
Statement-A : PTH hormone is a antagonistic hormone of thyrocalcitonin in reference of blood phosphate level.
Statement-B : Insulin is a steriod hormone which play a major role in regulation of glucose homoeostasis.
Statement-C : Glucocoticoids increase the break down of protein and increase blood glucose level by amino acid.

(1) All statements are incorrect
(2) Statement A is incorrect and Statement B & C are correct.q
(3) Statement A & B are incorrect and Statement C is correct.
(4) Statement B & C are incorrect and Statement A is correct.

67) Which statement is not true about steroid hormones

(1) They include hormones such as estrogen
(2) They do not bind to cell surface receptors
(3) The hormone receptor complex can not enter the nucleus
(4) Steroid hormones act slower than non steroid hormones

68) Match the columns and choose the correct option :

	Column-I	**Column-II**
(1)	Thyroxine	regulation of blood calcium
(2)	Cortisol	Fat synthesis
(3)	Insulin	Hypoglycemia
(4)	Melatonin	Suppression of immune system

(1) 1
(2) 2
(3) 3

(4) 4

69) Read the following statements and choose the correct option :

(1) Glucagon is a steroid hormone

(2) Cortisol produces anti inflammatory reactions

(3) Aldosterone suppresses the immune response

(4) Cortisol mainly helps in reabsorption of Na^+

70) Match column I and II and choose the correct option-

	Column-I		**Column-II**
(a)	Glucagon	(i)	Steroid
(b)	Epinephrine	(ii)	Peptide
(c)	Thyroxine	(iii)	Iodothyronine
(d)	Estrogen	(iv)	Amino acid derivative

(1) a-ii, b-iv, c-iii, d-i

(2) a-ii, b-iii, c-i, d-iv

(3) a-iv, b-ii, c-iii, d-i

(4) a-ii, b-iv, c-i, d-iii

71) Select the mismatch w.r.t hormone and its site of action.

(1) Melanocyte stimulating hormone - Skin cells

(2) Adrenocorticotrophic hormone - Adrenal cortex

(3) Follicle stimulating hormone - Ovary

(4) Gonadotrophin releasing hormone - Gonads

72) How many of the following statement are not incorrect?
(1) In human, pars intermedia is almost merged into pars nervosa.
(2) Anterior pituitary controls adrenal medulla gland through ACTH hormone.
(3) Posterior pituitary gland synthesizes & secretes 2 hormones.
(4) Pineal gland is not functional after puberty.
(5) Hypothalamic neurons secrete releasing & inihibitory hormones.

(1) 1

(2) 3

(3) 4

(4) 2

73) Read the following features in a baby and choose the disorder in which they are well observed ?
a. Abnormal skin
b. Low intelligence quotient
c. Deaf-mutism

d. Stunted growth
e. Mental retardation

(1) Cretinism
(2) Grave's disease
(3) Exophthalmic goitre
(4) Simple goitre

74) Read the following statements.
A : Invertebrates possess very simple endocrine system.
B : A large number of chemicals act as hormones and provide coordination in the vertebrates.
Choose the correct option.

(1) Both the statements are incorrect
(2) Only statement A is incorrect
(3) Only statement B is incorrect
(4) Both the statements are correct

75) Mark the correct matching regarding the hormone chemical nature and function :-

	Hormone	**Chemical nature**	**Function**
(1)	Gonadotrophins	Steroid	Development of follicle and produce antiinflammatory reactions
(2)	Parathormone	protein	Inhibit bone resorption to decrease blood Ca
(3)	Somatostatin	Peptide	Stimulate secretion of GH from adenohypophysis
(4)	Cortisol	Steroid	Stimulate gluconeogenesis lipolysis and proteolysis

(1) 1
(2) 2
(3) 3
(4) 4

76) Which type of mouth parts are found in Cockroach?

(1) Bitting and chewing
(2) Piercing
(3) Sucking
(4) Drilling

77) Out of following which match is incorrect for cockroach ?

(1) Crop-used for storage of food
(2) Gizzard-helps in grinding the food particles
(3) Hepatic caeca - at junction of midgut and hindgut
(4) Malpighian tubules - 100 to 150 in number

78) Malpighian tubules remove excretory products from

(1) Haemolymph

(2) Alimentary canal

(3) Both (1) and (2)

(4) None of these

79) Match the column-I & II with reference to cockroach and select the correct option :-

Column-I		Column-II	
(a)	Storing of food	(i)	Hepatic caeca
(b)	Grinding of food particle	(ii)	Proventriculus
(c)	Anal cerci	(iii)	Crop
(d)	Present at junction of foregut & midgut	(iv)	Malpighian tubules
(e)	Present at junction of midgut & Hindgut	(v)	10th segment

(1) (a)–(iv), (b)–(iii), (c)–(v), (d)–(ii), (e)–(i)

(2) (a)–(i), (b)–(iii), (c)–(v), (d)–(ii), (e)–(iv)

(3) (a)–(ii), (b)–(iv), (c)–(iii), (d)–(i), (e)–(v)

(4) (a)–(iii), (b)–(ii), (c)–(v), (d)–(i), (e)–(iv)

80) Match the following with reference to cockroach and select the correct option :

Column-I		Column-II	
(A)	Ovarioles	(i)	External genitals
(B)	Spermatophore	(ii)	Bundle of sperms
(C)	Phallomere	(iii)	Contain chain of developing ova
(D)	Gonopore	(iv)	Opening of ejaculatory duct

Options :

(1) A–(iii), B–(iv), C–(ii), D–(i)

(2) A–(iii), B–(ii), C–(i), D–(iv)

(3) A–(iv), B–(iii), C–(ii), D–(i)

(4) A–(ii), B–(iv), C–(iii), D–(i)

81) Brood or genital pouch in female cockroach is made by :-

(1) 7th sternum

(2) 7th sternum, 8th & 9th tergum

(3) 7th, 8th and 9th tergum

(4) 7th, 8th & 9th sternum

82) Read the following statements carefully :

(A) Visual units that compose the compound eyes of insects are called ommatidia.
(B) Night vision of cockroach is more sensitive but less resolution.
(C) Haemolymph enters heart through ostia and is pump from anterior to posterior direction.
(D) Haemolymph has no role in transportation of nutrients and waste.
How many above statements are correct

(1) Four
(2) Three
(3) Two
(4) One

83) Which of the following statement is incorrect w.r.t cockroach ?

(1) Hind wings are transparent, membranous and are used in flight.
(2) Gizzard has inner layer of thick circular muscles and thick outer layer of cuticle.
(3) In both sexes, the 10th segment of abdomen bears a pair of jointed filamentous structure called anal cerci.
(4) In males genital chamber bounded dorsally by 9th & 10th terga and ventrally by 9th sternum.

84) Read the following statements wrt. to cockroach:
(A) Respiratory system consist of a network of trachea that open through 10 pairs of small holes called spiracles most of them present laterally.
(B) Oesophagus opens into a sac like structure called crop used for storing of food.
(C) A ring of 3–4 pairs blind tubules present at the junction of Foregut & midgut.
(D) Malpighian tubules helps in removal of excretory products from haemolymph
How many above statements are correct.

(1) One
(2) Two
(3) Three
(4) Four

85) Which of the following statement is true for cockroach :-

(1) They are ureotelic
(2) Anal style are absent in females
(3) The larval stage is called tadpole
(4) The total number of ovarioles are eight in both ovaries.

86) _____ forms a diffusion boundary in our body

(1) Compound epithelium
(2) Simple columnar epithelium
(3) Simple squamous epithelium
(4) Simple cuboidal epithelium

87) It serves as a support framework for epithelium. It contains fibroblast, macrophages and mast

cells. It is known as.

(1) Dense regular tissue
(2) Adipose tissue
(3) Dense irregular tissue
(4) Areolar tissue

88) Distribution of tissue in our epidermis of skin is –

(1) Simple epithelial tissue
(2) Compound epithelial tissue
(3) Muscular tissue
(4) Connective tissue

89)

Cartilage present at tip of nose :-

(1) Elastic cartilage
(2) Hyaline cartilage
(3) Calcified cartilage
(4) White fibrous cartilage

90) Mark the mismatch pair :

(1) Unicellular glandular epithelium - goblet cells
(2) Endocrine glands - ducted glands
(3) Compound epithelium - protection function.
(4) Multicellular glands - Salivary glands

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	2	3	2	2	2	3	2	3	2	1	1	3	2	1	1	1	1	4	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	3	1	4	1	4	3	4	4	1	3	1	4	3	2	3	3	3	2	4
Q.	41	42	43	44	45															
A.	4	1	1	2	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	3	4	3	4	2	1	3	2	3	1	3	2	2	3	4	3	1	1	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	3	4	2	4	2	2	2	3	3	4	4	2	2	1	1	3	3	2	2
Q.	86	87	88	89	90															
A.	2	3	2	1	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	1	4	4	3	1	2	2	1	1	3	4	1	3	3	2	3	1	2	4	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	4	3	1	1	2	4	1	3	1	2	1	2	4	1	2	1	3	3	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	1	1	2	1	1	4	2	2	2	3	1	3	4	1	3	2	1	3	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	4	1	3	2	3	3	3	2	1	4	1	1	4	4	1	3	1	4	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	3	2	3	2	3	4	2	1	2										

SOLUTIONS

PHYSICS

1)

Explain the Question : For a body of **mass m = 2 kg** initially at rest, subjected to the **time-varying force (F−t) graph** shown, **find its final velocity v_2 after t = 15 sec**.
Concept : This question is based on the impulse - change in its linear momentum.
Formula : Impulse (J) = $\Delta p = m(v_2 - v_1)$ = Area of the F−t graph
Solution : Impulse = Δp = Area of force-time graph

$$m(v_2 - v_1) = \frac{1}{2} \times 2 \times 10 + 3 \times 10 + \frac{1}{2} \times 5 \times 10 + 10 \times 5 + 2 \times 10 + \frac{1}{2} \times 3 \times 10$$

$2(v - 0) = 10 + 30 + 25 + 50 + 20 + 15$
$v = 75$ m/sec
Final answer : 1

2) **Explain Question :** A force of **F Newton** produces an acceleration of **$1 m/s^2$** in a body. We need to determine the mass of the body.

Concept : This problem is based on **Newton's Second Law of Motion**.

Formula :
$F = ma$

Solution :

Given: $a = 1 m/s^2$ $F = \sqrt{(3)^2 + (4)^2 + (12)^2} = \sqrt{169} = 13$ N so, $m = \frac{F}{a}$

$$m = \frac{13}{1} = 13 \text{ kg}$$

Final Answer : Option (2)

3)

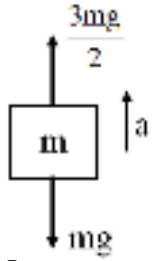
Explain the Question : We need to find the horizontal velocity of the block shown in the figure, from given data.
Concept : This question is based on constraint motion.
Solution :
$v' \sin\theta = v$

$$v' = \frac{v}{\sin\theta}$$

Final answer : 3

4) **Explain the question :** To find the acceleration of the block.
Concept: This question is based on motion of bodies connected with pulley systems without spring
Formula: Net force = mass × acceleration

Solution:



$$\left[\frac{3mg}{2} - mg\right] = ma$$

$\Rightarrow a = \left(\frac{g}{2}\right)$ upward

Final answer : 2

6) **Explain the Question :** We need to find the minimum force F, which is sufficient to keep the block in the state of equilibrium, against the rough vertical wall shown.

Concept : This question is based on block against rough vertical wall.

Solution :

$\mu N = Mg$

$0.5(F) = 5(10)$

$F = 100$ N

Final Answer : 2

7) $T_1 = \dfrac{(m_2 + m_3 + m_4)F}{m_1 + m_2 + m_3 + m_4}$

$\therefore T_1 = \dfrac{3}{4}F$

Similarly, $T_2 = \dfrac{(m_3 + m_4)F}{m_1 + m_2 + m_3 + m_4}$

or $T_2 = \dfrac{1}{2}F$

$T_3 = \dfrac{(m_4)F}{m_1 + m_2 + m_3 + m_4}$

or $T_3 = \dfrac{1}{4}F$

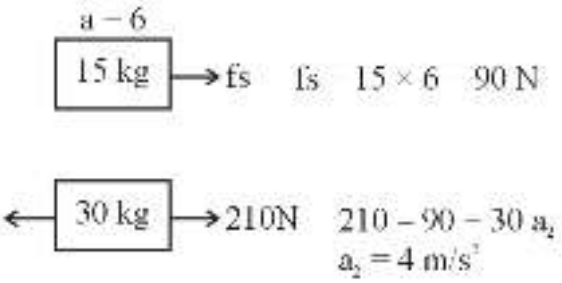
8) $a = \dfrac{200 - 16g}{16} = \dfrac{5}{2}$ m/sec^2

$T_P = 9(g + a)$

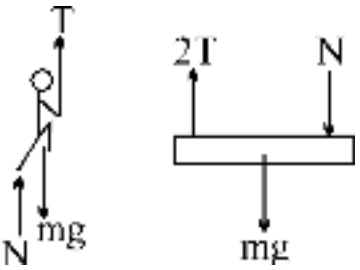
$T_P = 9\left(10 + \dfrac{5}{2}\right)$

$T_P = 112.5$ N

9)



10)



T + N = mg ...(1)

& 2T = N + mg ...(2)

From (1) & (2)

T = 250 N

11) Limiting friction between block and slab = $\mu_s m_A g$

= 0.6 × 10 × 10 = 60 N

But applied force on block A is 100 N. So the block will slip over a slab.

Now kinetic friction works between block and slab $F_k = \mu_k m_A g = 0.4 \times 10 \times 10 = 40$ N

This kinetic friction helps to move the slab

∴ Acceleration of slab = $\frac{40}{m_B} = \frac{40}{40} = 1$ m/s^2

12) $N_1 = 2a$, $N_2 = 1a$ $\left(a = \frac{3}{3} = 1 m/s^2\right)$

$\frac{N_1}{N_2} = \frac{2}{1}$

13)

Asking About:

Relationship between $\sqrt{E_k}$ and momentum (p).

Concept:

Kinetic energy is related to momentum

Formula

$E_k = \frac{p^2}{2m}$

Solution/Explanation/Calculation:

• Taking square root: $\sqrt{E_k} = \frac{p}{\sqrt{2m}}$
• This is a linear equation (y–mx), meaning the graph is a straight line passing through the origin

Final Answer :
3 (Straight line increasing)

14) **Question Explanation:**
A mass m slides from rest along a smooth hemispherical surface of radius R. We need to find its velocity at the lowest point.
Concept Based:
Mechanical energy is conserved.
Formula Used:
$mgh - \frac{1}{2}mv^2$
h = R (vertical drop from top to bottom of hemisphere)
Calculation:
Initial potential energy = mgR
Final kinetic energy = $\frac{1}{2}mv^2$
Equation: $mgR = \frac{1}{2}mv^2$
$v = \sqrt{2gR}$

15) **Question Explanation:** A particle is moved in a gravitational field of mass *M* from point *A* to point *B* via three different paths. We need to compare the work done along each path.

Concept Based:
The work done in a gravitational field depends only on the initial and final positions of the object, not the path taken. This is a result of the conservative nature of gravitational force.

3. Formula Used:
The work done by the gravitational force is given by: $W = -\Delta U = -mg\Delta h$ Where *U* is the gravitational potential energy, *m* is the mass of the particle, *g* is the gravitational acceleration, and Δ*h* is the change in height.

4. Calculation:
Since gravitational force is conservative, the work done in moving the particle from point *A* to point *B* depends only on the difference in height between these points, not the path taken. Therefore, regardless of the path, the work done is the same if the initial and final positions are the same.
Thus, we conclude:
$W_1 = W_2 = W_3$

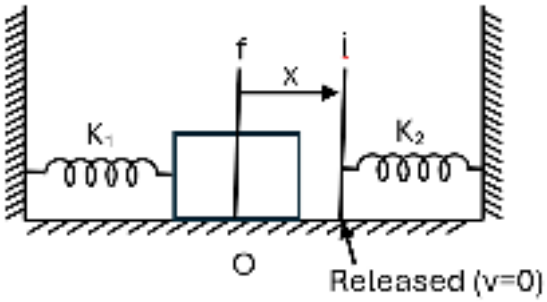
Final Answer:
Option 1: $W_1 = W_2 = W_3$.

16) **Explanation :**

Speed of the block at the mean position after release.

Concept :
Conservation of Mechanical Energy

Solution :

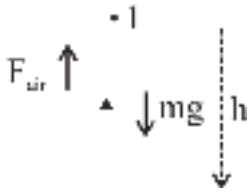


Potential energy stores in spring converts entirely to kinetic energy at mean position
$E_i = E_f$
$p_i + k_i = p_f + k_f$

$$\frac{1}{2}K_1x^2 + \frac{1}{2}K_2x^2 + 0 = 0 + \frac{1}{2}mv^2$$

$$V = \sqrt{\frac{(K_1 + K_2)x^2}{m}}$$

Final Answer :
Option (1)
Final Answer :
Option (1)



17)

$K_1 = 0$; $K_2 = \frac{1}{2} \cdot \left(\frac{2}{1000}\right) .30^2 = 0.9\,J$

$W_g = mgh = \frac{2}{1000} \times 10 \times (0.5 \times 1000) = 10\ J$

$W_{air} = ?$

According to WET,

$W_g + W_{air} = K_2 - K_1$

$10\ J + W_{air} = 0.9 - 0$

$W_{air} = -9.1\ J$

18) **Question Explanation**
A body of mass m moves along the x-axis from rest under a constant power P. We need to determine the relationship between velocity v and time f

Concept

Solution:

• Power in terms of velocity and mass

P = mav (P = F.V and F =Ma)

• Acceleration is the time derivative of velocity:

$$a = \frac{dv}{dt}$$

• Substituting is into the power equation.

$$P = mv\frac{dv}{dt}$$

• Rearranging

$$\frac{P}{m}dt = vdv$$

• Integrating both sides

$$\int vdv = \int \frac{P}{m}dt$$

• Performing integration:

$$\frac{v^2}{2} = \frac{P}{m}t$$

• Solving for v

$$v = \sqrt{\frac{2P}{m}t}$$

• This shows that $\propto t^{1/2}$

Correct Answer: Option (1) $\propto t^{1/2}$

Question Level: Tough

19) **Explain the question :** In this question, we need to find how much mass of water (M) a 2 Kw pump can lift to a height of 10 m in 1 minute.

Concept : This question is based on power.

Solution : Power, $P = \frac{\text{work done}}{\text{time}} = \frac{MgH}{t}$

$$\Rightarrow M = \frac{P \times t}{gH}$$

Given here

P = 2kW = 2000 watt, t = 1 min = 60 sec

g = 10 m/s^2, H = 10 M

$$\Rightarrow M = \frac{2000 \times 60}{10 \times 10} = 1200 \text{ kg} \approx 1200 \text{ litre} \qquad (1 \text{ litre} \approx 1 \text{ kg})$$

Final answer : Option (4)

20) **Concept Based:**

Relation between kinetic energy and momentum.

Formula Used:

$$K = \frac{p^2}{2m}$$

If p becomes 2p (100% increase), then

$$K' = \frac{(2p)^2}{2m} = \frac{4p^2}{2m} = 4K$$

Calculation:
The new kinetic energy is 4 times the initial kinetic energy, meaning an increase of
$\frac{K' - K}{K} \times 100 = \frac{4K - K}{K} \times 100 = 300\%$

Final Answer:
The correct option is (3).

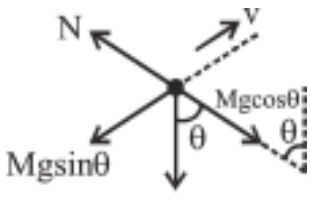
21)

$W_{friction} + W_{gravity} = \Delta KE$
$-mg\square + mgh - mgh' = 0$
$-\mu\square + h = h'$
$-4 + 5 = h'$
$h' = 1$ m

22) $W = Fd\cos 0°$
$= (1)(7) = 7J$

23) $V_{max} = \sqrt{\left(\frac{\tan\theta + \mu}{1 - \mu\tan\theta}\right) Rg}$
$= \sqrt{\left(\frac{\tan 45 + 0.5}{1 - 0.5\tan 45}\right) \times 1000 \times 9.8} = 172$ m/s

24) Since the maximum tension T_B in the string moving in the vertical circle is at the bottom and minimum tension T_T is at the top.
$\square\ T_B = \frac{mv_B^2}{L} + mg$ and $T_T = \frac{mv_T^2}{L} - mg$
$\square\ \frac{T_B}{T_T} = \frac{\frac{mv_B^2}{L} + mg}{\frac{mv_T^2}{L} - mg} = \frac{4}{1}$ or $\frac{v_B^2 + gL}{v_T^2 - gL} = \frac{4}{1}$
or $v_B^2 + gL = 4v_T^2 - 4gL$ but $v_B^2 = v_T^2 + 4gL$
$\square\ v_T^2 + 4gL + gL = 4v_T^2 - 4gL \Rightarrow 3v_T^2 = 9gL$
$\square\ v_T^2 = 3 \times g \times L = 3 \times 10 \times \frac{10}{3}$ or $v_T = 10$ m/sec

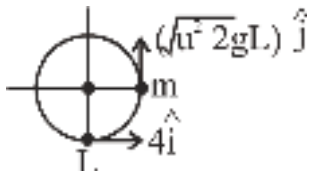


25) $Mg\cos\theta - N = \frac{Mv^2}{R}$

$N = Mg\cos\theta - \frac{Mv^2}{R}$

$\theta \downarrow \quad \cos\theta \uparrow \quad \Rightarrow N \uparrow$

26)



$|\vec{v}_m - \vec{u}|$

$= \left|\sqrt{u^2-2gL}\,\hat{j} - 4\hat{i}\right|$

$= \sqrt{\left(\sqrt{u^2-2gL}\right)^2 + 4^2}$

$= \sqrt{2\,(u^2-gL)}$

27) $\vec{a}_c = \frac{v^2}{r}\left(-\cos\theta\,\hat{i} - \sin\theta\hat{j}\right)$

$= \frac{2^2}{4}\left(-\cos 45\hat{i} - \sin 45\hat{j}\right)$

$= -\left(\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$

28) Velocity at the bottom = $\sqrt{2gh}$

For completing the loop

$\sqrt{2gh} = \sqrt{5gR}$

$\Rightarrow \quad R = \frac{2h}{5} = 2 \times \frac{5}{5} = 2cm$

29) **Concept :** Circular motion

Formula:

$a_c = \frac{V^2}{R}$

$a_T = R\alpha$

Mathematical calculation:

• If partical perform circular motion than if must have a_c but if speed of partical increase or decrease tan it have a_t

such that (c, d) are correct.

Correct answe with option : 4

Estimated Time : 1-2 minutes

Level : Easy

30)

For position B

$mg - N = \frac{mV_B^2}{R}$

for value of R

When N = 0 $mg \geq \frac{mV_B^2}{R}$

$R \geq \frac{V_B^2}{g}$ and $mg(h_A - h_B) = \frac{1}{2} mV_B^2$

$R \geq \frac{2g(h_A - h_B)}{g}$

$R \geq \frac{2g(25-15)}{g}$

$R \geq 2g$

$R_{min.} = 20m$

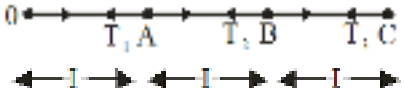
31)

32) $v^2 = \mu Rg$

or $v^2 = 0.8 \times 84.5 \times 10$

or $v = 26\ ms^{-1}$

33) Let ω is the angular speed of revolution.



$T_3 = m\omega^2 3l$

$T_2 - T_3 = m\omega^2 2l \Rightarrow T_2 = m\omega^2 5l$

$T_1 - T_2 = m\omega^2 2l \Rightarrow T_1 = m\omega^2 6l$

$T_3 : T_2 : T_1 = 3 : 5 : 6$

34)

$\omega = \frac{14}{25}$ rev/s $= \frac{14}{25} \times 2\pi$ rad/s^2

$a_c = \omega^2 r$

35)

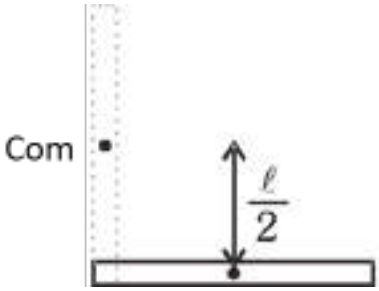
Explain Question:

Find work done by force to take a rod of M and l from horizontal to vertical position.

Concept:

Work done.

Solution:



Work done = mgh

Work done $= mg\frac{\ell}{2}$

Final answer: option (2)

36) $\vec{v}_1 = 3\hat{i} + 2\hat{j},\ \ \vec{v}_2 = -\hat{1} - 4\hat{j}$

All three fragments have equal masses, say 'm' each,

Then, by COLM

$m\vec{v}_3 + m\vec{v}_2 + m\vec{v}_1 = 0$

$\Rightarrow \vec{v}_3 = -(\vec{v}_2 + \vec{v}_1)$

$= -\left[-\hat{i} - 4\hat{j} + 3\hat{i} + 2\hat{j}\right]$

$\Rightarrow \vec{v}_3 = -2\hat{i} + 2\hat{j}$

37)

Momentum of skater A = 30 × 1 = 30 kgm/s

Momentum of skater B = 20 × 2 = 40 kgm/s

They are at right angles to each other.

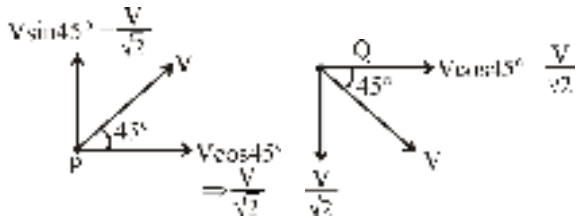
Resultant momentum = p

$\therefore (p)^2 = (30)^2 + (40)^2 = 900 + 1600 = 2500$

or p = 50 kgm/s

$\therefore$ Final velocity = $\frac{p}{\text{Total mass}}$

$= \frac{50}{(30+20)} = \frac{50}{50} = 1$ m/s



38)

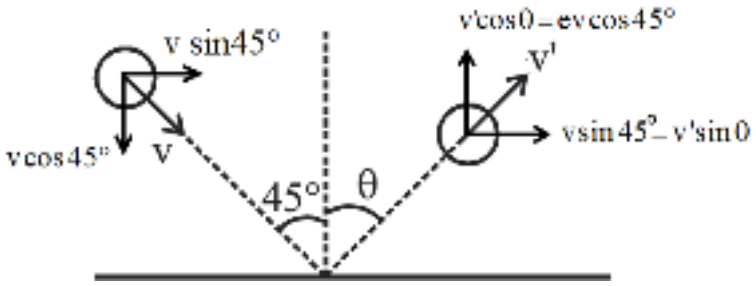
$\Delta V = \left(\frac{V}{\sqrt{2}}\right) - \left(-\frac{V}{\sqrt{2}}\right) \Rightarrow \sqrt{2}V$

Change in momentum $\Rightarrow \sqrt{2}mV$

39) **Question Explanation :** A ball hits a surface with velocity vv at a 45° angle and rebounds with velocity v′ at an angle θ. Given the coefficient of restitution e = $\frac{1}{\sqrt{3}}$, find the rebound angle after the collision.

Concept: Oblique collision - Coefficient of restitution

Solution:



During collision between ball and surface velocity in horizontal direction remain same.

$v \sin 45° = v' \sin\theta$ ...(1)

$ev \sin 45° = v' \cos\theta$ ...(2)

Equation (1) ÷ Equation (2)

$\frac{1}{e}\tan 45° = \tan\theta \quad \therefore \tan 45° = 1; \quad \therefore e = \frac{1}{\sqrt{3}}$

$\frac{1}{\frac{1}{\sqrt{3}}} = \tan\theta = \sqrt{3}$

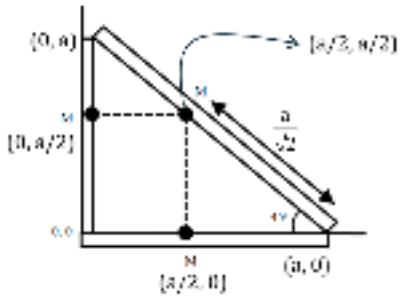
$\therefore \theta = 60º$

Final answer: Option (2)

40) **Explain Question :** As shown in figure find Co-ordinates of center of mass.

Concept: Center of mass.

Solution:



$$X_{com} = \frac{m_1x_1 + m_2x_2 + m_3x_3}{M_1 + M_2 + M_3}$$

$$X_{com} = \frac{m\frac{a}{2} + m\frac{a}{2} + m \times 0}{m + m + m} = \frac{a}{3}$$

Similarly, $Y_{com} = \frac{a}{3}$

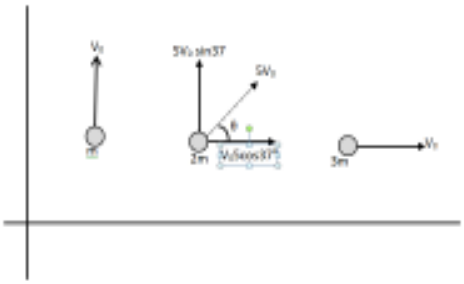
$r_{com} = \left(\frac{a}{3}, \frac{a}{3}\right)$

Final answer: Option (4)

41) **Explanation:**
As shown in figure find velocity of center of mass.

Concept:
Motion of center of mass.

Solution:



Formula:

$$\vec{V}com = \frac{m_1\vec{v}_1 m_2\vec{v}_2 + m_3\vec{v}_3}{m_1 + m_2 + m_3}$$

$$\vec{v}_2 = 4v_0\hat{i} + 3v_0\hat{j}$$

$$\vec{v}_2 = v_0\hat{j}$$

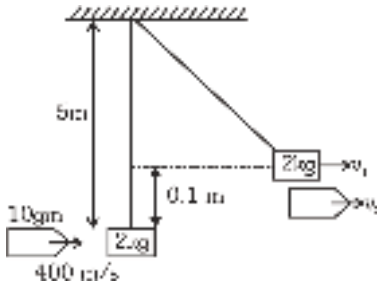
$$\vec{v}_3 = 4v_0\hat{i}$$

$$\vec{V}com = \frac{mv_0\hat{j} + 2m(3v_0\hat{j} + 4v_0\hat{j} + 3mv_0\hat{i})}{m + 2m + 3m}$$

$$\vec{V}com = \frac{11v_0}{6}(\hat{i}) + \frac{7v_0}{6}(\hat{j})$$

Final answer: Option:(D)

42)



Applying momentum conservation

$$\frac{10}{1000} \times 400 + 0 = 2 \times v_1 + \frac{10}{1000} \times v_2$$

$\Rightarrow \quad 4 = 2v_1 + 0.01v_2 \quad .....(1)$

Applying work energy theorem for block

$W = \Delta KE$

$$\Rightarrow \quad 2 \times 10 \times 0.1 = \frac{1}{2} \times 2 \times v_1^2$$

$\Rightarrow \quad v_1 = \sqrt{2} = 1.4$ m/s

Putting the value of v_1 in equation (1)

$4 = 2 \times 1.4 + 0.01\ v_2 \Rightarrow v_2 = 120$ m/s

43) **Question Explanation:** Two masses, m and 3m, are connected by a string in an Atwood machine. When released, we need to find the acceleration of the center of mass (COM).
Concept: Motion of center of mass (COM)
The center of mass accelerates as the total external force (due to the difference in gravitational forces) acts on the combined system of masses.
Solution:
Acceleration of each block is

$$a = \frac{(3m-m)}{3m+m}g = \frac{g}{2}$$

$$\vec{a}_{cm} = \frac{3m\vec{a}_1 + m\vec{a}_2}{3m+m}$$

Both mass have same magnitude of acceleration but in opposite direction $\vec{a}_1 = -\vec{a}_2 = a$ Let)

$$a_{cm} = \left(\frac{3m-m}{4m}\right) \times \frac{g}{2} = \frac{g}{4}$$

Final answer : option (1)

44) **Explanation:**
This question based on c.o.m. position or it lies side of the body or not.

Concept :
Center of mass

Solution:
About of C.O.M.

$$m_1\vec{r}_1 + m1\vec{r}_2 = 0$$

$$m_1\vec{r}_1 = m_2\vec{r}_2$$

$$m_1 > m_2$$

$$r_2 > r_1$$

C.O.M. of a body can be lie outside of the body (ex.ring)

Final answer: Option(B)

45) Momentum is conserved.

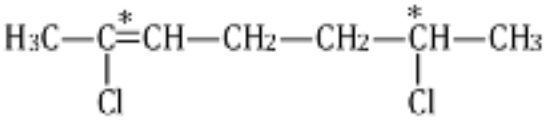
CHEMISTRY

46)

NCERT Based

47)

NCERT Based



48)

Unsymmetrical compound

So No. of G.I. = 2^n where n → no. of stereogenic center

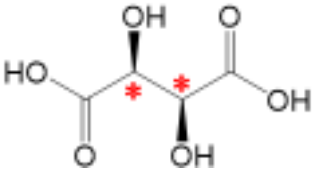
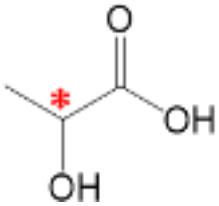
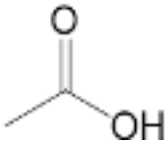
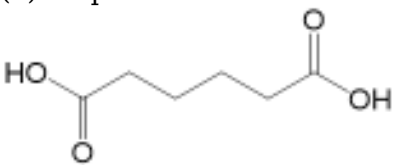
$= 2^2 = 4$

49)

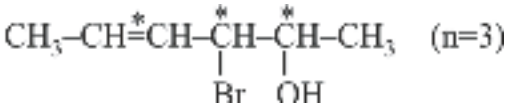
Explain Question: Select the chiral molecules.

Concept: Chirality of molecule.

Solution:

Given compounds	POS	COS	Chiral / achiral molecule
(a) d-Tartaric acid 	(□)	□	Chiral molecule
(b) Lactic acid 	(□)	□	Chiral molecule
(c) Acetic acid 	□	□	Achiral molecule
(d) Adipic acid 	□	□	Achiral molecule

Final answer : 3



50)

SI = 2^n

51)

NCERT Based

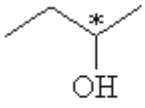
52)

NCERT Based

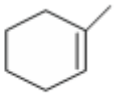


53)
All 4 groups attached to carbon is different so it is chiral compound.

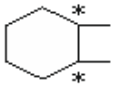
54)



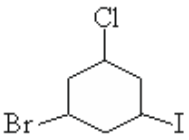
n = 1



NIL



n = 2



n = 3

55)

NCERT Based

56)

NCERT Based

57) a, b and c

58)

NCERT Based

59)

Center of symmetry

60)

2-butanol

61)

NCERT Based

62)

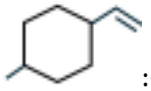
Fact

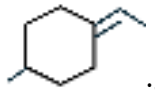
63) **Explanation:**
Heat of hydrogenation order of given alkene.

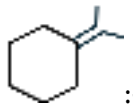
Concept:

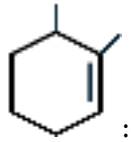
Heat of hydrogenation $\propto \frac{1}{\text{stability of alkene}}$

Solution:

[P]  :- ∝ – C – H ⇒ 1

[Q]  :- ∝ – C – H ⇒ 7

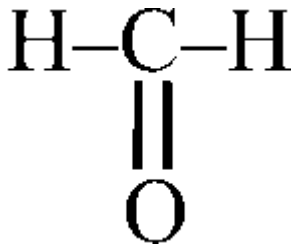
[R]  :- ∝ – C – H ⇒ 10

[S]  :- ∝ – C – H ⇒ 6
 stability of Alkene is :- R > Q > S > P
 order HOH is :- P > S > Q > R.

Final Answer:
Option (1)

64)

Explain the Question : The problem asks us to evaluate the truth of the given Assertion and Reason regarding heterolytic fission and determine the correct relationship between them.
Concept : This question is based on Bond fission, Polar covalent bonds, Electron distribution
Solution :
Assertion : "Heterolytic fission occurs readily in polar covalent bonds." → This is **true**, because in polar covalent bonds, one atom is more electronegative and already pulls electron density, making it easier for the bond to break unevenly.
Reason : "Heterolytic fission involves breaking of bonds in such a way that the shared pair of electrons go with one atom." → This is also **true**, as by definition heterolytic cleavage results in a cation and an anion.
The Reason correctly explains why heterolytic fission is favored in polar covalent bonds (electrons are already biased toward the more electronegative atom).
Final answer : 1



65)
NCERT based

66)

H_r involved due to resonance.

67)

E, F and G are resonating structures

68)

NCERT

69)

Fact

70)

NCERT Based

71)

NCERT based

72)

NCERT based

73)

– NO_2 opperate both –I and – M in X.

74)

NCERT Based

75)

NCERT Based

76)

(D) $NO_2^- + \text{sil}\ H_2S^{H^\oplus}O_4 \rightarrow H^\oplus + NO_2^- \rightarrow HNO_2$

$HNO_2 \xrightarrow{H^+} HNO_3 + No$

No NO_2 = brown $\xrightarrow{air}$

$\rightarrow No_3^\ominus$ + conc $H_2\ SO_4 \rightarrow H^\oplus + No_3^\ominus \rightarrow HNO_3$

$HNO_3 \xrightarrow[Cu]{H^+} NO_2 + H_2O$

brown gas

77) $Ni^{+2} + 2DMG \rightarrow \underset{\text{Rosy red}}{[Ni(DMG)_2]}$

78) **Explanation :-** Question is asking about the compound When a mixture of solid NaCl, solid $K_2Cr_2O_7$ is heated with concentrated H_2SO_4, orange red vapours are obtained
Concept - This question is based on Colors of vapours), $K_2Cr_2O_7$ properties
Solution :- Chromyl chloride produce orange red vapours When a mixture of solid NaCl, solid $K_2Cr_2O_7$ is heated with concentrated (H_2SO_4)
Final Answer 2

79) **Question Explanation**
Question is asking about the condition when during salt analysis basic radicals, precipitate formation will take place,
Concept –
This concept based on basic radicals, precipitate. **Solution –** During salt analysis of basic radicals, a precipitate will form when the ionic product of the salt exceeds its solubility product. This means that the concentration of the ions in solution is greater than what they can hold when they form a solid, causing the solid to come out of solution as a precipitate.
Final Answer (2)

80) **Question Explanation**
Question is asking about reddish brown vapour
Concept –
This concept based on Colours of vapour **Solution –**
When a mixture containing bromides is heated with concentrated sulfuric acid (H_2SO_4) and manganese dioxide (MnO_2), reddish-brown vapours of bromine (Br_2) are liberated. This reaction is a common lab method for producing bromine
Final Answer (1)

81) **Question Explanation**
Question is asking about the element that gives green flame in flame test

Concept
This Concept based on Flame test
Solution
Barium salts produce a bright green flame, often described as yellow-green, apple-green, or lime-green
Final Answer (1)

82)

Explaining the Question :-
Question is asking about the group reagent for group II
Concept:-
group reagent of basic radicals
Solution :-
H_2S in presence of HCl is the group reagent for group II
Final Answer :- 3

83)

$2CuSO_4 + K_4[Fe(CN_6)] \rightarrow Cu_2[Fe(CN)_6] \downarrow + 2K_2SO_4$
(Chocolate brown)

84)

Explaining the Question :-

Question is asking about the anion of the gas which gives rotten egg like smell

Concept:-

Odour of gas

Solution :-

Salt (S^{2-}) + 2HCl → H_2S (gas) + other products

The gas produced is **hydrogen sulphide** (H_2S), which is responsible for the characteristic odor of rotten eggs.

Final Answer :- 2

85)

Explaining the Question :-

Question is asking about incorrect match of identification test and colour

Concept:-

identification test and colour

Solution :-

Chromyl chloride ($CrCl_3$) is **red or pink** in color instead of green hence this match is incorrect
Final Answer :- 2

86)

Explaining the Question :-

Question is asking about matching of radicals with their group

Concept:-

group is asking about matching of radicals with their group

Solution :-

Pb^{+2} — I group

Fe^{+3} — III group

Ni^{+2} — IV group

Mg^{+2} — VI group
Final Answer :- 2

89)

Generated by Allie

Problem Statement: The question asks us to match four different molecules with their characteristic odors or appearances. We have a list of molecules (CO2, I2, H2S, Br2) and a list of descriptions of smells or visible vapors. The task is to correctly pair each molecule with the odor or characteristic associated with it.

Underlying Concept: The problem involves understanding the distinctive physical and chemical properties of common substances, mainly focusing on their odors and visible characteristics. Certain molecules have well-known smells or colors of fumes that help identify them. For example, hydrogen sulfide (H2S) is known for its rotten egg smell, iodine (I2) produces violet fumes, bromine (Br2) appears as reddish brown vapors, and carbon dioxide (CO2) is odorless.

Tips and Tricks: Memorize key characteristic odors and appearances: H2S has a rotten egg smell, I2 vapors are violet, Br2 is reddish-brown, and CO2 is odorless. Associating each molecule with its unique sensory property helps quickly solve such matching questions.

Common Mistakes: Confusing the odors of H2S and CO2 or mixing the colors of iodine and bromine vapors can lead to incorrect matching. Remember that CO2 is odorless, not smelly, which can be a common misconception.

Why Other Options Are Incorrect?: Incorrect options might switch the odors and colors of iodine and bromine or wrongly assign a smell to carbon dioxide. Such mistakes ignore the known properties of these molecules. For example, volcanic or rotten egg smell is characteristic of H2S, not CO2.

90)

(A) Lake Test → Al^{+3}
(B) Nessler's Reagent → NH_4^+
(C) Potassium sulphocyanide → Fe^{+3}
(D) Brown ring Test → NO_3^-

BIOLOGY

91)

Module.135

92)

XIth NCERT Old, Pg. No. - 232

93) NCERT Page No. 157

94) **Explaining :** RQ = CO_2 released / O_2 consumed. For glucose: 6/6 = 1.
Concept :
Solution : $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$
Final Answer : (3)

95) Sol. [a] No. of carbon atoms in following intermediates- [α-ketoglutaric acid (5C), Oxaloacetic acid (4C), Citric acid (6C), Succinic acid (4C)].

96) Sol. [b] Diagramatic presentation of ATP synthesis in mitochondria.

97) Sol. [b] The correct sequence of electron transporters in ATP synthesis is cytochrome b, cytochrome c1, cytochrome c cytochrome a, and cytochrome a3.

98) The initial step of the Tricarboxylic Acid cycle involves the condensation of an acetyl group with oxaloacetic acid to form citric acid. This reaction is catalyzed by the enzyme citrate synthase.

99) NCERT Pg. # 228

100)

Explain Question: The question asks which outcome correctly describes a product or feature of glycolysis.
Concept: This question is based on Glycolysis
Solution:
Glycolysis produces 2 pyruvic acid molecules per glucose.
No O_2 used, no CO_2 released, and NADH (not $NADPH_2$) is formed.

Final answer: 3.

101) NCERT (XIth) Pg. # 230, 231

102) NCERT XI, (E) Pg. # 107

103)

NCERT (XIth) Pg. # 237

104) Module Pg. # 205

105)

NCERT (XI) Pg # 231
(ANS. 2,3)

106)

NCERT (XI) Pg. # 235

107) NCERT Pg # 157,164

108) NCERT XI, Pg. # 232

109) **Explain Question :** question is asking which enzyme(s) are involved in the conversion of pyruvic acid into ethyl alcohol during fermentation

Concept : Respiration in Plants – Anaerobic Respiration (Fermentation)

Solution : in yeast and some plant tissues, pyruvic acid formed during glycolysis is converted into ethyl alcohol under anaerobic conditions.
Pyruvic acid → Acetaldehyde + CO_2
  Enzyme: Pyruvate decarboxylase
Acetaldehyde → Ethyl alcohol
  Enzyme: Alcohol dehydrogenase

Final answer : Option (4)

110)

NCERT XI Pg. # 232

111) NCERT (XI) Pg. # 230

112) NCERT (XIth) Pg. (E) # 231 (Topic : Aerobic respiration)

113) It acts as a source of energy

114) The appearance of different forms of leaves on the same plant species

115) Simple organic substances of different chemical composition

116) 2, 4-D and NAA

117) Ripening

118) Controlling mobilisation of nutrients

119) Dedifferentiation

120) A–*Gibberella fujikuroi,* B–Gibberellins

121) Auxins and Cytokinins

122) 3 NADH2, 1 FADH2

123) 12

124) a(iii), b(iv), c(i), d(ii)

125) NCERT (XI) Pg. # 249,250 para-4,5, line2,3,4,5

126)

NCERT PAGE NO. 170

127)

NCERT Pg. 241

128)

NCERT-XI, Pg. # 243

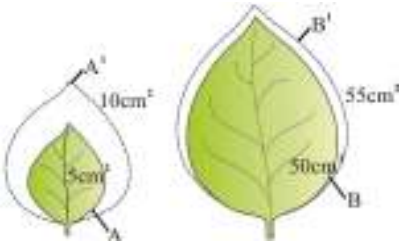
129) NCERT Pg # 241

130)

Ncert page 176.

131) NCERT Pg. # 247

132) **Question Explanation :**
To find the wrong statement with respect to leaf A and B in the given diagram.



Concept : Plant Growth
Solution :
For A leaf :
Absolute growth : 10–5 = $5cm^2$
Relative growth : 10-5/5 = $1cm^2$
For B leaf
Absolute growth : 55-50 = 5 cm^2
Relative growth : 50-50/50 = 1/10 cm^2
So, the incorrect statement is : 1. Relative growth rate of A is twice as that of B.
Final Answer : Option (1)

133) NCERT Pg. # 170

134)

NCERT-XI, Pg.# 248, 249, 250

135)

The correct answer is 1. A = Initial slow growth, B = Rapid growth, C = Phase of growth during limited nutrient supply.
Here's a breakdown of the sigmoid growth curve and the phases indicated:

A. **A - Initial Slow Growth (Lag Phase):** This is the initial phase where growth is slow. It represents a period of adaptation and preparation. The organism is adjusting to its environment, synthesizing necessary enzymes and materials before rapid growth can occur.
B. **B - Rapid Growth (Log/Exponential Phase):** In this phase, growth accelerates rapidly. Resources are abundant, and conditions are optimal for growth and reproduction. The curve becomes steep and almost vertical, indicating the exponential increase in size, number, or biomass.
C. **C - Phase of Growth During Limited Nutrient Supply (Stationary Phase):** As resources become limited, growth slows down and eventually plateaus. The birth/growth rate becomes equal to the death/decay rate. The curve flattens out, indicating that the population or organism has reached its carrying capacity. Nutrient availability, space constraints, and accumulation of waste products contribute to limiting further growth.

136) **Question Explanation**
Select the incorrect statement w.r.t. acromegaly.

Explanation:
acromegaly is caused by hypersecretion (not hyposecretion) of growth hormone (GH) from the pituitary gland in adults. This leads to abnormal growth of bones, especially in the face, hands, and feet.

Final Answer (1)

137)

The correct answer is 4. Only statement B is correct.

Explanation:
Statement A is incorrect: While the adrenal glands do secrete small amounts of weak androgens (like DHEA) that contribute to the growth of axial and pubic hair during puberty, these hormones are produced by the **adrenal cortex**, not the adrenal medulla.

Statement B is correct: Cortisol, a glucocorticoid produced by the adrenal cortex, plays a vital role in maintaining the cardiovascular system as well as kidney function.

138)

The correct answer is **2. Oxytocin**

Explanation:
The features described are characteristic of oxytocin:
• (a) Released by neurohypophysis.
• (b) Acts on the smooth muscles of uterus and stimulates their contractions during parturition.
• (c) Acts on the mammary gland for the ejection of milk.

139)

The correct answer is **2. Influence the male sexual behaviour**

Solution:
Androgens, primarily testosterone are key hormones responsible for the development of male secondary sexual characteristics and influence libido and other male sexual behaviours.

140)

The correct answer is **2. Pituitary gland**
Solution:
The sella turcica is a saddle-shaped depression in the shenoid bone at the base of the skull that primarily lodges the pituitary gland.

141)

Question Explanation
Assertion (A): Exopthalmic goitre is a disease characterised by enlargement of the parathyroid gland, increased BMR, weight loss, protrusion of eyeballs, etc.
Reason (R): Exophtalmic goitre is a form of hyperparathyroidism.
In the light of above statements, select the **correct** option.

Explanation
Assertion (A):
Exophthalmic goitre is a disease characterized by enlargement of the parathyroid gland, increased BMR, weight loss, protrusion of eyeballs, etc.
This statement is incorrect because exophthalmic goitre (or Graves' disease) is caused by hyperthyroidism, which involves the thyroid gland, not the parathyroid gland. It results in symptoms like increased BMR, weight loss, and protrusion of eyeballs. The parathyroid gland is not involved in exophthalmic goitre.
Reason (R):
Exophthalmic goitre is a form of hyperparathyroidism.This statement is also incorrect. Exophthalmic goitre is a form of hyperthyroidism, not hyperparathyroidism. Hyperparathyroidism is a condition related to excessive secretion of parathyroid hormone (PTH), which leads to calcium imbalances, not the symptoms described in exophthalmic goitre.

Final Answer (3)

142)

The correct answer is:
Option (1) - Hyperglycemia, Diabetes Mellitus

Explanation:

A. **Insulin** is a hormone secreted by the **pancreas** that helps lower blood sugar levels by promoting glucose uptake into cells.

B. If insulin is **malfunctioning or deficient**, glucose cannot enter the cells effectively, leading to **high blood sugar** (**hyperglycemia**).

This condition leads to **Diabetes Mellitus**, which is characterized by chronic high blood sugar.

143)

Correct Answer:
Option (3)
A-ii, B-iii, C-iv, D-i

Solution :

- A. Parafollicular cells → (ii) Calcitonin hormone
- B. Parathyroid gland → (iii) Collips hormone
- C. Thyroid gland → (iv) Tetra iodothyronin
- D. Zona glomerulosa → (i) Aldosterone

144)

NCERT-XI; Page No. # 241

145)

NCERT-XI; Page No. # 244-245

146)

NCERT-XI; Page No. # 246

147)

NCERT-XI; Page No. # 245

148)

NCERT-XI; Page No. # 245

149)

NCERT-XI; Page No. # 246

150)

NCERT-XI; Page No. # 242-243

151)

NCERT-XI; Page No. # 244

152)

NCERT-XI; Page No. # 241

153)

NCERT-XI; Page No. # 240

154)

NCERT-XI; Page No. # 247-248

155)

NCERT-XI; Page No. # 242

156)

NCERT-XI; Page No. # 241, 243, 245

157)

NCERT-XI; Page No. # 241

158) NCERT, Pg # 334,336,337

159) NCERT, Pg # 336,337

160)

NCERT Page No. 337, 338

161) NCERT Page no 333, 334

162)

NCERT-XI; Page No. # 240

163) NCERT Page no 335

164) NCERT Page no 331

165) NCERT XI, Pg. # 335

166)

NCERT Pg. # 112

167)

Question Explanation: Identifying incorrect match for cockroach parts.
Concept: Cockroach anatomy
Solution: Hepatic caecae are located at the junction of the foregut and midgut, not the midgut and hindgut. The other options correctly describe the functions of the crop, gizzard, and the number of Malpighian tubules.
Final Answer: Hepatic caeca - at junction of midgut and hindgut

168)

Haemolymph

169)

Question Explanation: Matching cockroach parts with their functions.
Concept: Cockroach anatomy
Solution: (a) Storing food: Crop (iii). (b) Grinding food: Proventriculus (ii). (c) Anal cerci are on the 10th segment (v). (d) Junction of foregut & midgut has Hepatic caeca (i). (e) Junction of midgut & hindgut has Malpighian tubules (iv).
Final Answer: (a)-(iii), (b)-(ii), (c)-(v), (d)-(i), (e)-(iv)

170) NCERT-XI, Pg. # 114

171) NCERT-XI, Pg. # 112

172) NCERT-XI, Pg. # 113,114

173) NCERT-XI, Pg. # 112,113

174)

Question Explanation: Identifying correct statements about cockroach anatomy.
Concept: Cockroach anatomy and physiology
Solution: Statement (A) is correct; cockroaches have 10 pairs of spiracles. Statement (B) is also correct; the oesophagus opens into the crop for food storage. Statement (C) is incorrect; there are 6-8 pairs of hepatic caecae, not 3-4. Statement (D) is correct; Malpighian tubules are the excretory organs.
Final Answer: Three

175) NCERT-XI, Pg. # 112,114

176)

Question Explanation: **:** Identify the epithelial type that facilitates diffusion across body surfaces.

Concept : This question is based on Epithelial tissue.

Solution : The correct answer is option 3 (Simple squamous epithelium) because
* It consists of thin , flat cells with minimal barrier, ideal for diffusion and filtration.
* Found in alveoli of lungs, Bowman's capsule, and capillary walls.
Final Answer : option (3)

177) **Areolar tissue :**
Support framework for epithelium; contains fibroblasts, macrophages, and mast cells.

178)

Compound epithelial tissue

179)

Question Explanation: The question is asking which type of cartilage is present at the tip of the nose.

Concept: Types of cartilage.

Solution : In the tip of the nose, the cartilage is flexible and contains elastic fibres, so it is classified as elastic cartilage in this context. Therefore, the correct answer is elastic cartilage.

Final Answer: Option (1)

180) Endocrine glands are ductless glands. Their products are secreted directly into fluid bathing the gland.
Whereas, exocrine glands are ducted glands. Their secretions are released through ducts or tubes.